End-to-End Decision-Focused Learning for Perishable Inventory-Routing under Asymmetric Spoilage and Stockout Costs

Abstract

Perishable supply chains require coordinated inventory replenishment and vehicle routing under uncertain demand and finite shelf lives. In a conventional "Predict-then-Optimize" (PTO) workflow, forecasts are trained using statistical losses such as mean squared error (MSE) and then passed to a mixed-integer linear program (MILP). This separation can misalign forecasts with operational costs: over-forecasting may increase spoilage, while under-forecasting may increase stockout costs. We study an end-to-end decision-focused learning (DFL) framework for the multi-period perishable inventory-routing problem (PIRP), with asymmetric spoilage and stockout penalties. We formulate a two-stage bilevel recourse model with LP duality to derive recourse subgradients with respect to replenishment decisions, and establish complete-recourse and feasibility guarantees under the stated assumptions. We compare established SPO+, interior-point differentiation (IntOpt), and differentiable perturbed optimizer (DPO) approaches, and define a custom stop-gradient surrogate for RHS demand along with a two-phase schedule that transitions from LP-relaxation training to MILP fine-tuning.

Across five random seeds and 1,500 training epochs, Two-Phase DFL achieves mean out-of-sample normalized decision regret of 22.28% (between-seed standard deviation 0.08%; instance-level standard deviation 18.77%) and mean total operational cost of $\$2,143.60$ (between-seed standard deviation \$1.33) on the clustered topology, compared with $\$1,873.58$ for the clairvoyant benchmark. On the primary benchmark at baseline capacity ($\gamma = 1.0$), paired comparisons with step-down Holm-Bonferroni adjustment indicate lower regret than each of the 13 baselines ($p_{\text{Holm}} < 0.05$). The method also achieves a 5.66× wall-clock speedup in a separate solver-profiling experiment (8.43 s vs. 47.71 s) and lower validation regret than direct MILP training. A factorial sensitivity study spanning 81 configurations and period- and store-level capacity diagnostics examines performance across balanced and slack fleet regimes ($\gamma \geq 1.0$). In the tested tight-capacity setting ($\gamma = 0.8$), LightGBM has lower regret than Two-Phase DFL (5.60% vs. 6.91%), indicating that the method's advantage is not uniform across capacity regimes.

Keywords: Decision-focused learning; Predict-then-optimize; Perishable inventory-routing; Asymmetric costs; Linear programming relaxation warmup; Smart Predict-then-Optimize (SPO+).

1 Introduction

In the distribution of perishable food items, pharmaceuticals, and temperature-sensitive goods, supply chain managers face the daunting challenge of coordinating multi-period inventory replenishment across distributed retail outlets while simultaneously executing capacitated vehicle delivery tours from a central depot [**?** **?** **?**]. Products in these sectors possess strictly finite shelf lives ($\tau \in \{2, 3, 5\}$ periods). Any inventory remaining unconsumed upon reaching its expiration age spoils and must be disposed of at substantial operational and environmental cost [**?** **?**]. Simultaneously, retail store stockouts incur punitive emergency replenishment costs or severe lost-sales penalties [**?** **?**].

Traditionally, supply chain management addresses this problem by decomposing it into two sequential, decoupled phases—a paradigm widely referred to as **Predict-then-Optimize (PTO)** [**?** **?**]:

(i) **Prediction Stage**: Statistical or machine learning models (e.g., Ridge regression, LightGBM [**?**], or Temporal Fusion Transformers [**?**]) are trained on historical sales, weather covariates, calendar events, and promotional data to forecast future demand by minimizing symmetric loss metrics such as Mean Squared Error (MSE) or Mean Absolute Error (MAE):

$$\mathcal{L}_{\text{MSE}}(\theta) = \frac{1}{NT} \sum_{i=1}^{N} \sum_{t=1}^{T} \left(\hat{d}_{i,t}(x_{i,t}; \theta) - d_{i,t} \right)^2 .$$

(ii) **Optimization Stage**: Point forecasts $\hat{d}_{i,t}$ are plugged directly into a deterministic Mixed-Integer Linear Program (MILP) representing the Multi-Period Perishable Inventory-Routing Problem (PIRP) to compute optimal delivery quantities and vehicle travel schedules [**?** **?**].

1.1 The Methodological Failure of Two-Stage PTO in Perishable Operations

Although standard in industrial software, the two-stage PTO paradigm suffers from a fundamental mathematical flaw in perishable logistics: **symmetric forecasting losses are decoupled from the asymmetric economic realities and combinatorial thresholds of the downstream operational problem** [**?** **?**]. Specifically:

- **Severe Cost Asymmetry**: In perishable supply chains, the cost consequences of forecast errors are highly asymmetric. If customer demand is over-forecast by 10 units, excess inventory is delivered; if it cannot be consumed within shelf-life τ, it triggers spoilage and waste penalties $p_s = 5h$ (where h denotes the unit inventory holding cost per period). Conversely, if demand is under-forecast by 10 units, retail store shelves stock out, triggering emergency expedited shipments or lost-sales penalties $p_u \in \{10h, 20h, 50h\}$. An under-forecast error can be 2 to 10 times more damaging financially than an over-forecast error of identical magnitude. Symmetric MSE penalizes $+10$ and -10 identically, failing to incentivize the predictive model to hedge against catastrophic stockouts.

- **Combinatorial Capacity Discontinuities**: Vehicle routing decisions are governed by discrete capacity thresholds [**?** **?**]. If cumulative deliveries on a tour remain within vehicle payload capacity Q, a small forecast error may cause little routing disruption. If the error pushes the required load above Q, the fleet planner may need an additional route, increasing travel costs. Standard statistical loss functions treat every unit of error uniformly, regardless of whether it crosses a routing threshold or falls within existing capacity slack.

As a result, statistical models with strong predictive accuracy (low MSE or MAE) can still yield substantial **Decision Regret** when their forecasts are used in distribution networks [**?** **?**].

1.2 Research Questions and Contributions

This paper bridges this fundamental gap by formulating and solving the **End-to-End Decision-Focused Learning (DFL) framework for Multi-Period Perishable Inventory-Routing**. We address three core research questions:

RQ1. How can combinatorial inventory-routing problems with *uncertain right-hand-side (RHS) constraints* be solved end-to-end via decision-focused learning through bilevel recourse formulations and exact dual subgradients without risking solver infeasibility?

RQ2. Which gradient approximation and surrogate loss mechanisms (SPO+, IntOpt, DPO) offer the highest solution quality and stability when backpropagating through NP-hard combinatorial MILP routing structures?

RQ3. How can we overcome the prohibitive training complexity of repeatedly solving MILPs over hundreds of neural network epochs without degrading final operational decision quality?

To answer these questions, this paper establishes four primary methodological contributions:

1. **Exact Linear Recourse Reformulation with Complete Feasibility**: Prior DFL literature (e.g., **? ? ?**]) predominantly assumes uncertainty appears exclusively in the linear objective function ($\min c^T w$). In inventory routing, however, uncertain demand appears in the right-hand-side inventory conservation constraints. We introduce an exact two-stage linear recourse reformulation with explicit emergency replenishment ($y^+_{it} \geq 0$), spoilage disposal ($y^-_{it} \geq 0$), and proactive excess disposal slacks ($\delta_{i,t,a} \geq 0$). We prove (**Theorem 1**) that the resulting first-stage feasibility set $\mathcal{W}_1$ is compact and demand-independent, while the second-stage recourse set $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ is feasible for every nonnegative demand vector under the stated store-capacity constraints. Thus, any nonnegative predicted-demand vector yields a feasible recourse problem under the stated model assumptions.

2. **Surrogate Gradient Approaches**: We compare an SPO+ baseline, RHS interior-point differentiation, and Gaussian perturbed optimization, and define a custom stop-gradient surrogate for RHS demand. The custom update is explicitly treated as a heuristic rather than an exact subgradient of the MILP task loss. Their assumptions and empirical performance are examined in Sections 4 and 6.

3. **Two-Phase Training Schedule (LP Warmup to MILP Fine-Tuning)**: Direct end-to-end training over MILP combinatorial solvers requires solving thousands of NP-hard branch-and-cut programs, creating a severe wall-clock bottleneck and causing noisy gradient updates in early epochs. We develop a formal two-phase training protocol (**Algorithm 2**): continuous Linear Programming (LP) relaxation training during early epochs ($E_{\text{trans}} = 75\%$) to smooth the polyhedral gradient landscape and rapidly learn coarse feature representations, followed by discrete MILP fine-tuning. We empirically substantiate this two-phase schedule by measuring rapid validation-regret convergence and documenting order-of-magnitude wall-clock speedups (5.66× solver speedup) over direct MILP training.

4. **Benchmarking across Models and Spatial Topologies**: We compare 10 two-stage PTO baselines with four end-to-end DFL methods across Solomon topologies (`C1`, `R1`, `RC1`), shelf lives ($\tau \in \{2, 3, 5\}$), and stockout-to-holding cost ratios ($p_u/h \in \{10, 20, 50\}$). On the primary benchmark at baseline capacity ($\gamma = 1.0$), across five seeds and 1,500 training epochs, the proposed method achieves 22.28% mean regret (between-seed standard deviation 0.08%; instance-level standard deviation 18.77%) and mean total cost of $\$2,143.60$ (between-seed standard deviation $\$1.33$). Paired comparisons with step-down Holm-Bonferroni adjustment find lower regret than each of the 13 baselines ($p_{\text{Holm}} < 0.05$). We also report sensitivity results in which LightGBM has lower regret than Two-Phase DFL in the tested tight-capacity

setting ($\gamma = 0.8$; 5.60% vs. 6.91%); this is an empirical result for the tested configuration, not a general capacity boundary.

The remainder of this manuscript is structured as follows. Section 2 reviews related literature and positions our work. Section 3 details the multi-period PIRP mathematical model, FIFO inventory dynamics, and proves Theorem 1 for the two-stage bilevel recourse reformulation. Section 4 derives the end-to-end DFL surrogate losses (SPO+, IntOpt, DPO) and specifies Algorithm 2 for two-phase LP-warmup training. Section 5 outlines the empirical dataset, spatial topologies, baselines, and evaluation metrics. Section 6 presents benchmark performance, speedup profiles, sensitivity analysis, and managerial insights. Section 8 concludes.

2 Literature Review and Methodological Positioning

This research integrates three vibrant streams of literature: (i) Predict-then-Optimize and Decision-Focused Learning; (ii) Contextual optimization with uncertain constraints; and (iii) Perishable inventory-routing problems.

2.1 Predict-then-Optimize and Decision-Focused Learning

The integration of machine learning predictions directly into mathematical programming models has received significant attention across operations research and computer science [**?** **?**]. In their seminal work, **?**] established the Smart Predict-then-Optimize (SPO) framework, proving that training prediction models against downstream decision regret outperforms standard MSE-based two-stage estimation. To overcome the non-differentiability of combinatorial argmin operators, they proposed the SPO+ surrogate loss, demonstrating its Fisher consistency and convexity.

Concurrently, researchers developed continuous relaxation and implicit differentiation techniques. **?**] introduced OptNet, differentiating through quadratic programs via the Karush-Kuhn-Tucker (KKT) conditions. **?**] extended this to general disciplined convex programs. For combinatorial problems with discrete decision spaces, **?**] smoothed integer linear programs using quadratic regularization. **?**] advanced this concept by embedding interior point barrier solvers (IntOpt), differentiating the homogeneous self-dual embedding of linear programs. In a parallel direction, **?**] and **?**] introduced stochastic perturbation methods, showing that injecting random noise into linear optimizers produces smooth, non-zero gradients via Fenchel-Young conjugates. Recently, **?**] unified several of these methodologies into the open-source PyEPO library, while **?**] provided an extensive empirical survey of DFL techniques across standard combinatorial benchmarks (shortest path, knapsack, and bipartite matching).

However, virtually all aforementioned DFL literature focuses on problems where the unknown contextual parameter appears exclusively in the *objective function coefficients* ($c \in \mathbb{R}^m$). In core supply chain problems, demand uncertainty inevitably resides in the *constraints*, creating potential infeasibility when applying standard DFL techniques.

2.2 Contextual Optimization under Constraint and RHS Uncertainty

Extending decision-focused learning to right-hand-side (RHS) constraint uncertainty is an emerging and challenging frontier [**?**]. When predicted parameters govern feasibility sets, small estimation errors can render optimization models strictly infeasible, causing standard automatic differentiation engines to fail.

Non-parametric approaches to contextual optimization include the prescriptive analytics framework of **?**], who used kernel regression, random forests, and k-nearest neighbors to re-weight historical scenarios in Sample Average Approximation (SAA). **?**] demonstrated how regularized empirical risk minimization can solve newsvendor problems with contextual features. More recently, **?**] investigated surrogate losses for contextual linear optimization

with robust constraints (SPO-RC), deriving conditions under which surrogate losses maintain consistency under uncertainty sets. While these works address constraint uncertainty, they either assume static single-period settings or require extensive multi-scenario sampling that becomes computationally prohibitive when coupled with NP-hard vehicle routing constraints.

2.3 Perishable Inventory-Routing Problems (PIRP)

The inventory-routing problem (IRP) integrates inventory management and vehicle routing over a discrete multi-period planning horizon [**?** **?**]. In classic IRPs, products have infinite shelf lives. However, in cold chains and food distribution, perishability introduces temporal decay dynamics [**?** **?**].

?] presented an early foundational model for allocating perishable items with uncertain demands. **?**] formulated exact branch-and-cut algorithms for the multi-period PIRP with deterministic demand, tracking inventory age categories under First-In, First-Out (FIFO) and Last-In, First-Out (LIFO) policies. **?**] extended perishable inventory routing to stochastic settings, using dynamic programming and branch-and-cut to handle uncertain customer consumption. **?**] developed advanced polyhedral formulations and hybrid heuristics for large-scale PIRP instances. **?**] and **?**] incorporated environmental emissions and perishability into closed-loop green supply chain designs.

Despite these significant advancements, existing PIRP literature remains divided into two distinct extremes: either deterministic models that rely on point forecasts generated by unaligned two-stage statistical models, or stochastic/robust programming formulations that suffer from the curse of dimensionality and cannot ingest rich high-dimensional exogenous covariates (e.g., store promotions, weather series, and calendar interactions). Our work bridges this divide by enabling deep neural networks to learn directly from high-dimensional covariates through end-to-end backpropagation against the combinatorial PIRP objective.

2.4 Methodological Positioning

Table 1 systematically compares this paper against leading works in the literature across key operational and methodological dimensions.

Table 1: Methodological Positioning: Comparing Our Work against Relevant Literature.

Study	**Horizon**	**Routing Fleet**	**Perishable Shelf-Life**	**Uncertainty Locus**	**Cost Asymmetry**	**Optimization Paradigm**	**Learning Paradigm**	**Training Scheme**
?]	Multi	Capacitated	Discrete τ	None (Det.)	Symmetric	Branch-and-Cut	None	Offline Exact
?]	Multi	Single	Discrete τ	RHS Demands	Symmetric	Stochastic DP	None	Scenario Tree
?]	Multi	Capacitated	Discrete τ	None (Det.)	Symmetric	Branch-and-Cut	None	Offline Exact
?]	Multi	Capacitated	Continuous	RHS Demands	Symmetric	Chance-Constrained	None	Heuristic
?]	Single	None	None	RHS Demands	Asymmetric	SAA Weights	Local Trees/kNN	Two-Stage
?]	Single	None	Single-period	RHS Demands	Asymmetric	Linear Program	ERM Splines	Two-Stage
?]	Single	Shortest Path	None	Objective c	General	Linear Program	SPO+ Surrogate	Direct Solver
?]	Single	Bipartite	None	Objective c	General	QP Relaxation	Task Loss	Direct Solver
?]	Single	Knapsack/SP	None	Objective c	General	LP (HSD)	IntOpt Barrier	Direct Solver
?]	Single	Ranking/SP	None	Objective c	General	Perturbed Opt.	Fenchel-Young	Monte Carlo
?]	Single	None	None	RHS Robust	Violations	Robust LP	SPO-RC+	Direct Solver
This Work (DFL-PIRP)	**Multi** ($T \geq 5$)	**Capacitated** ($K \geq 2, Q$)	**Discrete** τ ($\tau \in \{2,3,5\}$)	**RHS Demands Bilevel Recourse**	**Severe Asym.** ($p_u \gg p_s$)	**Recourse MILP FIFO Dynamics**	**End-to-End DFL (SPO+/IntOpt/DPO)**	**Two-Phase (LP Warmup)**

As highlighted in Table 1, to our knowledge, this paper is among the first to study end-to-end decision-focused learning for a multi-period, capacitated inventory-routing problem with discrete perishability tracking, asymmetric operational costs, and a two-phase training schedule.

3 Mathematical Formulation and Bilevel Recourse Reformulation

3.1 Sets, Indices, Physical Parameters, and Notation Table

Table 2 defines the unified mathematical notation, sets, parameters, first-stage decision variables, and second-stage recourse variables governing the Multi-Period Perishable Inventory-Routing Problem (PIRP).

Table 2: Comprehensive Mathematical Notation for the Multi-Period Perishable Inventory-Routing Problem (PIRP).

Category	Symbol	Description / Definition	Range / Baseline	Unit
Sets	$\mathcal{V}$	Complete set of network nodes $\{0\} \cup \mathcal{N}$, where 0 is the depot	$\lvert\mathcal{V}\rvert = N + 1 = 11$	Nodes
	$\mathcal{N}$	Set of geographically dispersed retail stores (customer outlets)	$\{1, 2, \ldots, N\}, N = 10$	Stores
	$\mathcal{E}$	Set of directed travel arcs $\{(i, j) \in \mathcal{V} \times \mathcal{V} : i \neq j\}$	$\lvert\mathcal{E}\rvert = 110$	Arcs
	$\mathcal{T}$	Discrete multi-period planning horizon	$\{1, 2, \ldots, T\}, T = 5$	Periods / Days
	$\mathcal{K}$	Homogeneous delivery vehicle fleet stationed at the central depot	$\{1, \ldots, K\}, K = 2$	Vehicles
	$\mathcal{A}$	Perishable inventory age cohorts tracked at retail stores	$\{1, \ldots, \tau\}, \tau \in \{2, 3, 5\}$	Age cohorts
Indices	i, j	Node indices ($i = 0$ for depot, $i, j \in \mathcal{N}$ for customer retail stores)	$i, j \in \mathcal{V}$	Dimensionless
	t	Planning period index across the multi-period horizon	$t \in \mathcal{T} = \{1, \ldots, T\}$	Period
	k	Individual delivery vehicle index in the fleet	$k \in \mathcal{K} = \{1, \ldots, K\}$	Vehicle
	a	Product age cohort index ($a = 1$ fresh arrival, $a = \tau$ expiration)	$a \in \mathcal{A} = \{1, \ldots, \tau\}$	Period
Parameters	N	Number of customer retail stores in the distribution network	$N = 10$	Stores
	T	Planning horizon length	$T = 5$	Periods / Days
	K	Number of delivery vehicles available in the homogeneous fleet	$K = 2$	Vehicles
	Q	Payload capacity per vehicle ($K \cdot Q = 200$ total fleet capacity)	$Q = 100$ ($Q \in \{80, 100, 150\}$)	Units
	C_i	Maximum storage capacity of retail customer store $i \in \mathcal{N}$	$C_i = 70.0$	Units
	τ	Maximum product shelf life before unconsumed stock spoils	$\tau \in \{2, 3, 5\}$ (baseline $\tau = 3$)	Periods
	h	Baseline holding cost for fresh stock ($a = 1$)	$h = 1.0$	\$/unit/period
	ϵ	Marginal holding cost aging surcharge per period ($h_a = h + (a-1)\epsilon$)	$\epsilon = 0.01$	\$/unit/period
	h_a	Age-differentiated holding cost for stock of age $a \in \{1, \ldots, \tau - 1\}$	$0 < h_1 < \cdots < h_{\tau-1} < p_s$	\$/unit/period
	p_s	Unit waste penalty for expired / spoiled inventory	$p_s = 5.0 = 5h$	\$/unit
	p_u	Unit stockout penalty for emergency replenishment	$p_u \in \{10, 20, 50\}h$	\$/unit
	c_{ij}	Transportation cost along directed arc $(i, j) \in \mathcal{E}$	Solomon Euclidean distance	\$
	$\tilde{d}_{it}$	Realized customer demand at retail store i in period t	M5-calibrated series	Units
	$I_{i,0,a}$	Initial inventory of age a at store i entering the horizon	$I_{i,0,a} = 0.0, \forall i, a$	Units
First-Stage Decisions	x^k_{ijt}	Binary routing variable: 1 if vehicle k traverses arc (i, j) in period t, 0 else	$x^k_{ijt} \in \{0, 1\}$	Binary
	u^k_{it}	MTZ continuous tour order auxiliary variable for vehicle k at store i in period t	$u^k_{it} \in [1, N]$	Dimensionless
	q^k_{it}	Quantity of fresh product delivered to store i by vehicle k in period t	$q^k_{it} \geq 0$	Units
	q_{it}	Total aggregate fresh replenishment delivered to store i in period t	$q_{it} = \sum_{k \in \mathcal{K}} q^k_{it}$	Units
Second-Stage Recourse	$I_{i,t,a}$	Carryover inventory on hand of age a at store i at the end of period t	$I_{i,t,a} \geq 0, a \in \{1, \ldots, \tau - 1\}$	Units
	$w_{i,t,a}$	Quantity of inventory of age a consumed to meet demand at store i in period t	$w_{i,t,a} \geq 0, a \in \{1, \ldots, \tau\}$	Units
	y^+_{it}	Emergency replenishment recourse (unmet customer demand / stockout)	$y^+_{it} \geq 0$	Units
	y^-_{it}	Total spoiled and proactively disposed inventory at store i in period t	$y^-_{it} \geq 0$	Units
	$\delta_{i,t,a}$	Proactive excess disposal slack of age a purged into total disposal y^-_{it}	$\delta_{i,t,a} \geq 0, a \in \{1, \ldots, \tau\}$	Units

Within-Period Operational Sequence of Events: In each planning period $t \in \mathcal{T}$ at each retail store $i \in \mathcal{N}$, physical operations proceed through an explicit 5-step sequence:

Step 1. Replenishment Delivery Arrival and Aging: Delivery vehicles arrive from the central depot carrying fresh products of age $a = 1$, providing incoming supply $S_{i,t,1} = q_{it} = \sum_{k \in \mathcal{K}} q^k_{it}$. Simultaneously, carryover inventory from period $t - 1$ advances in age by one period: $S_{i,t,a} = I_{i,t-1,a-1}$ for all cohorts $a \in \{2, \ldots, \tau\}$.

Step 2. Customer Demand Realization: Customer demand $\tilde{d}_{i,t} \geq 0$ is realized on store shelves.

Step 3. FIFO Inventory Depletion: Customer demand is satisfied from available stock ($S_{i,t,a}$) following the First-In, First-Out (FIFO) oldest-first discipline ($w_{i,t,\tau} \to w_{i,t,\tau-1} \to \cdots \to w_{i,t,1}$). If total on-hand stock is insufficient ($\sum_{a=1}^{\tau} S_{i,t,a} < \tilde{d}_{i,t}$), an emergency replenishment recourse $y_{it}^{+} = \tilde{d}_{i,t} - \sum_{a=1}^{\tau} w_{i,t,a} > 0$ is triggered at unit stockout penalty p_u.

Step 4. Expiration and Proactive Excess Disposal: Unconsumed stock reaching expiration age τ cannot be retained and is discarded as terminal spoilage ($S_{i,t,\tau} - w_{i,t,\tau} \geq 0$). Additionally, if total surviving inventory exceeds store physical capacity C_i, proactive disposal slacks $\delta_{i,t,a} \geq 0$ purge excess units into total disposal $y_{it}^{-} = \sum_{a=1}^{\tau} \delta_{i,t,a}$, incurring unit waste penalty p_s.

Step 5. Carryover Storage and Holding Cost Assessment: The surviving non-expired inventory is carried forward to period $t+1$ as end-of-period inventory $I_{i,t,a} = S_{i,t,a} - w_{i,t,a} - \delta_{i,t,a} \geq 0$, subject to store storage capacity $\sum_{a=1}^{\tau-1} I_{i,t,a} \leq C_i$ and incurring age-differentiated holding costs $h_a I_{i,t,a}$.

Age-Differentiated Holding Costs: To reflect quality degradation, shelf inspection labor, dynamic markdown overhead, and elevated spoilage risk, retail inventory incurs **age-differentiated holding costs**:

$$0 < h_1 < h_2 < \cdots < h_{\tau-1} < p_s, \tag{1}$$

parameterized as $h_a = h + (a-1)\epsilon$ for $a \in \{1, \ldots, \tau-1\}$, where $h = 1.0$ is the baseline holding cost for fresh stock and $\epsilon = 0.01$ is the marginal aging surcharge per period. Goods that reach age τ during period t cannot be carried forward to period $t+1$; any unconsumed quantity spoils, incurring unit waste penalty $p_s = 5h = 5.0$. If customer demand exceeds available on-hand stock, an emergency shipment or stockout penalty $p_u \in \{10h, 20h, 50h\}$ is incurred per unmet unit.

Uncapacitated Cross-Dock Central Depot Assumption: At the central distribution depot ($i = 0$), consistent with modern cold-chain cross-dock distribution operations [**? ?**], the depot operates as an **uncapacitated cross-dock supply source**. Upstream agricultural suppliers and processing plants deliver fresh products on demand at the start of each period t. Products arrive at age $a = 1$ and are directly cross-docked into outgoing delivery vehicles without requiring multi-period inventory state tracking or depot capacity constraints. Consequently, vehicle loading and dispatch at node 0 are unconstrained by central depot inventory dynamics.

3.2 Decision Variables

The optimization decisions over the multi-period planning horizon comprise:

- $x_{ijt}^{k} \in \{0, 1\}$: Binary arc traversal variable, equal to 1 if vehicle $k \in \mathcal{K}$ traverses arc $(i, j) \in \mathcal{E}$ in period $t \in \mathcal{T}$, and 0 otherwise.
- $u_{it}^{k} \in [1, N]$: Auxiliary continuous variable for Miller-Tucker-Zemlin (MTZ) subtour elimination representing the visit order of vehicle k at store $i \in \mathcal{N}$ in period t.
- $q_{it}^{k} \geq 0$: Quantity of fresh goods delivered to store $i \in \mathcal{N}$ by vehicle $k \in \mathcal{K}$ in period t. The total replenishment delivery to store i is $q_{it} = \sum_{k \in \mathcal{K}} q_{it}^{k}$.
- $I_{i,t,a} \geq 0$: Carryover inventory on hand of age $a \in \{1, \ldots, \tau-1\}$ at store $i \in \mathcal{N}$ at the end of period t.

- $w_{i,t,a} \geq 0$: Quantity of inventory of age $a \in \{1, \ldots, \tau\}$ consumed to fulfill customer demand at store i during period t.
- $y_{it}^{+} \geq 0$: Recourse variable for unmet demand (emergency replenishment / stockout) at store i in period t.
- $y_{it}^{-} \geq 0$: Recourse variable for total spoiled and proactively disposed inventory at store i at the end of period t.
- $\delta_{i,t,a} \geq 0$: Proactive disposal slack for cohort a into y_{it}^{-} to guarantee physical storage capacity feasibility.

3.3 FIFO Perishable Inventory Dynamics and Optimality Proof

In perishable food and cold-chain distribution, retail stores operate under a First-In, First-Out (FIFO) inventory policy: customers and store managers consume the oldest available inventory first to minimize waste [**?** **?**]. In this subsection, we formalize the age-tracked inventory flow equations, define the FIFO consumption discipline, prove that age-differentiated holding costs guarantee that every optimal solution strictly obeys FIFO, and specify the post-decision FIFO evaluation simulator.

3.3.1 Inventory Conservation and Balance Constraints

Let $S_{i,t,a}$ denote the total on-hand stock of age $a \in \mathcal{A} = \{1, \ldots, \tau\}$ available at store $i \in \mathcal{N}$ at the beginning of period $t \in \mathcal{T}$ (after vehicle deliveries arrive, but before customer demand is fulfilled):

$$S_{i,t,1} = q_{it} = \sum_{k \in \mathcal{K}} q_{it}^{k}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{2}$$

$$S_{i,t,a} = I_{i,t-1,a-1}, \quad \forall i \in \mathcal{N}, t \in \{2, \ldots, T\}, a \in \{2, \ldots, \tau\}, \tag{3}$$

with initial inventory $S_{i,1,a} = I_{i,0,a-1}$ for $t = 1$ and $a \in \{2, \ldots, \tau\}$.

Customer demand $d_{i,t}$ is fulfilled by consuming available stock across age cohorts ($w_{i,t,a}$), with any deficit covered by emergency recourse y_{it}^{+}:

$$\sum_{a=1}^{\tau} w_{i,t,a} + y_{it}^{+} = d_{i,t}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}. \tag{4}$$

Consumption from each cohort is bounded by available stock:

$$0 \leq w_{i,t,a} \leq S_{i,t,a}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, a \in \{1, \ldots, \tau\}. \tag{5}$$

End-of-period carryover inventory $I_{i,t,a}$ is updated as:

$$I_{i,t,1} = q_{it} - w_{i,t,1} - \delta_{i,t,1}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{6}$$

$$I_{i,t,a} = I_{i,t-1,a-1} - w_{i,t,a} - \delta_{i,t,a}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, a \in \{2, \ldots, \tau - 1\}, \tag{7}$$

where $\delta_{i,t,a} \geq 0$ represents proactive excess disposal slacks. For goods reaching the maximum shelf life τ, unconsumed stock cannot be carried over and must be discarded as spoilage:

$$\delta_{i,t,\tau} = S_{i,t,\tau} - w_{i,t,\tau} = I_{i,t-1,\tau-1} - w_{i,t,\tau}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{8}$$

$$y_{it}^{-} = \sum_{a=1}^{\tau} \delta_{i,t,a}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{9}$$

$$I_{i,t,\tau} \equiv 0, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}. \tag{10}$$

Finally, physical retail storage capacity limits cumulative carryover stock:

$$\sum_{a=1}^{\tau-1} I_{i,t,a} \leq C_i, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}. \tag{11}$$

3.3.2 Formal FIFO Consumption Discipline

Definition 1 (FIFO Consumption Discipline)**.** A consumption vector $\mathbf{w}_{i,t} = (w_{i,t,1}, \ldots, w_{i,t,\tau})$ satisfies the **First-In, First-Out (FIFO)** oldest-stock-first discipline if younger inventory cohorts are consumed only after all older available cohorts have been fully depleted:

$$\forall 1 \le a' < a \le \tau : \quad w_{i,t,a'} > 0 \implies w_{i,t,a} = S_{i,t,a}. \tag{12}$$

Equivalently, this priority rule can be expressed as the complementarity condition:

$$w_{i,t,a'} \cdot (S_{i,t,a} - w_{i,t,a}) = 0, \quad \forall 1 \le a' < a \le \tau. \tag{13}$$

Without age-differentiated holding costs, standard linear balance constraints (4)–(11) do not strictly enforce (13) because an LP solver may be indifferent between consuming intermediate cohorts of different ages when future demand is large enough to prevent spoilage. The following lemma proves that introducing strictly increasing holding costs $h_1 < h_2 < \cdots < h_{\tau-1} < p_s$ guarantees that every optimal solution automatically and uniquely satisfies the FIFO discipline.

3.3.3 Optimality Proof for FIFO Consumption

Lemma 1 (Optimal FIFO Consumption)**.** *Suppose unit holding costs are strictly age-increasing:* $0 < h_1 < h_2 < \cdots < h_{\tau-1}$*, the terminal spoilage penalty satisfies* $p_s > \sum_{a=1}^{\tau-1} h_a$*, and the unit stockout penalty satisfies* $p_u > \sum_{a=1}^{\tau-1} h_a$*. Let* $Z^* = (\mathbf{x}^*, \mathbf{u}^*, \mathbf{q}^*, \mathbf{I}^*, \mathbf{w}^*, \mathbf{y}^{+*}, \mathbf{y}^{-*}, \boldsymbol{\delta}^*)$ *be an optimal solution to the Multi-Period PIRP (either the discrete MILP or its continuous LP relaxation) without redundant proactive intermediate disposal (*$\delta^*_{i,t,k} = 0$ *for* $k < \tau$*). Then, there exists an optimal solution that strictly satisfies the FIFO consumption discipline* (12)*. Furthermore, in any operational setting where customer demand in subsequent periods is positive or under an arbitrarily small tie-breaking priority* $\varepsilon_{fifo} > 0$ *for fresher inventory retention, every optimal solution strictly satisfies FIFO.*

Proof. We establish the result by an exchange perturbation argument. Suppose Z^* is an optimal candidate solution that violates the FIFO discipline. By Definition 1, there exists at least one retailer $i \in \mathcal{N}$, period $t \in \mathcal{T}$, and a pair of age cohorts a' and a with $1 \le a' < a \le \tau$ such that:

$$w^*_{i,t,a'} > 0 \quad \text{and} \quad w^*_{i,t,a} < S^*_{i,t,a}.$$

That is, a positive quantity of the younger cohort a' was consumed while available stock of the older cohort a was left unconsumed.

Define the strictly positive perturbation quantity:

$$\delta = \min\left\{w^*_{i,t,a'},\ S^*_{i,t,a} - w^*_{i,t,a}\right\} > 0.$$

We construct an alternative candidate solution $\tilde{Z} = (\mathbf{x}^*, \mathbf{u}^*, \mathbf{q}^*, \tilde{\mathbf{I}}, \tilde{\mathbf{w}}, \tilde{\mathbf{y}}^+, \tilde{\mathbf{y}}^-, \tilde{\boldsymbol{\delta}})$ by shifting δ units of consumption from the younger cohort a' to the older cohort a during period t, leaving vehicle routing ($\mathbf{x}^*$), tour ordering ($\mathbf{u}^*$), and delivery quantities ($\mathbf{q}^*$) unchanged:

$$\begin{aligned}
\tilde{w}_{i,t,a} &= w^*_{i,t,a} + \delta \le S^*_{i,t,a}, \\
\tilde{w}_{i,t,a'} &= w^*_{i,t,a'} - \delta \ge 0, \\
\tilde{w}_{i,t,k} &= w^*_{i,t,k}, \quad \forall k \in \mathcal{A} \setminus \{a', a\}.
\end{aligned}$$

1. Feasibility Verification:

- **Demand Satisfaction**: Total period-t consumption is invariant:

$$\sum_{k=1}^{\tau} \tilde{w}_{i,t,k} = (w^*_{i,t,a} + \delta) + (w^*_{i,t,a'} - \delta) + \sum_{k \ne a,a'} w^*_{i,t,k} = \sum_{k=1}^{\tau} w^*_{i,t,k}.$$

Hence, $\tilde{y}^+_{it} = y^{+*}_{it}$, so stockout penalties and demand satisfaction (4) remain identical.

- **Store Capacity**: If $a < \tau$, end-of-period inventories change by $\tilde{I}_{i,t,a} = I^*_{i,t,a} - \delta$ and $\tilde{I}_{i,t,a'} = I^*_{i,t,a'} + \delta$. Total carried inventory is unchanged:

$$\sum_{k=1}^{\tau-1} \tilde{I}_{i,t,k} = (I^*_{i,t,a} - \delta) + (I^*_{i,t,a'} + \delta) + \sum_{k \neq a,a'} I^*_{i,t,k} = \sum_{k=1}^{\tau-1} I^*_{i,t,k} \leq C_i,$$

preserving store capacity (11). If $a = \tau$, consuming δ units of cohort τ reduces terminal spoilage by δ ($\tilde{y}^-_{it} = y^{-*}_{it} - \delta \geq 0$). Setting proactive disposal slack $\tilde{\delta}_{i,t,a'} = \delta$ if store capacity is binding preserves exact inventory balance and storage feasibility.

2. Cost Difference Analysis: We evaluate the net cost difference $\Delta\text{Cost} = \text{Cost}(\tilde{Z}) - \text{Cost}(Z^*)$. Since routing and replenishment are fixed, $\Delta\text{Cost}_{\text{route}} = 0$ and $\Delta\text{Cost}_{\text{stockout}} = 0$.

Case 1: $a < \tau$ (Both cohorts are intermediate). In period t, reducing inventory of cohort a by δ and increasing cohort a' by δ changes immediate holding cost by:

$$\Delta\text{Cost}_t = h_{a'}\delta - h_a\delta = -(h_a - h_{a'})\delta < 0,$$

which is strictly negative because $a' < a \implies h_{a'} < h_a$. In subsequent periods $s = t+1, \ldots, T$, the surviving inventory under $\tilde{Z}$ has age $a' + (s-t) < a + (s-t)$ and incurs strictly lower holding cost $h_{a'+s-t} < h_{a+s-t}$ while held. Furthermore, cohort a' has $\tau - a' > \tau - a$ periods of remaining shelf life, delaying or preventing spoilage. Hence $\Delta\text{Cost} < 0$.

Case 2: $a = \tau$ (Older cohort is at expiration age). In Z^*, unconsumed stock of age τ spoiled at period t, incurring $p_s\delta$. Under $\tilde{Z}$, consuming this expiring stock saves $-p_s\delta$. The δ unconsumed units of younger cohort $a' < \tau$ are carried forward with $\tau - a'$ remaining shelf-life periods. If future customer demand in periods $s \in \{t+1, \ldots, t+\tau-a'\}$ is positive, this surviving stock fulfills demand and eliminates an emergency stockout or displaces future delivery, saving $p_u\delta$. Because $p_u > \sum_{s=a'}^{\tau-1} h_s$, this net savings strictly exceeds cumulative intermediate holding costs:

$$\Delta\text{Cost} \leq -p_s\delta + \sum_{s=a'}^{\tau-1} h_s\delta - p_u\delta < -\left(p_s - \sum_{s=a'}^{\tau-1} h_s\right)\delta < 0.$$

In the edge case where all downstream customer demands across the remaining horizon are zero and $q^*_{it} = 0$, proactive disposal of cohort a' (or terminal disposal at age τ) incurs at most $p_s\delta$, yielding $\Delta\text{Cost} \leq -p_s\delta + p_s\delta = 0$.

Therefore, transforming any non-FIFO solution into a FIFO solution never increases total operational cost ($\Delta\text{Cost} \leq 0$), establishing the existence of an optimal solution satisfying FIFO. Furthermore, in all operational scenarios where subsequent customer demand is positive or intermediate holding applies, $\Delta\text{Cost} < 0$, establishing that every optimal solution strictly satisfies FIFO. □

Remark 1 (Breaking Simplex Basis Degeneracy). Under conventional uniform holding costs ($h_a = h$ for all $a \in \{1, \ldots, \tau-1\}$), intermediate unconsumed stock incurs identical objective coefficients regardless of age cohort. In the linear programming (LP) continuous relaxation, this creates an underdetermined subspace where the reduced costs for non-FIFO basic variables vanish ($\bar{c}_{i,t,a} = \bar{c}_{i,t,a'}$). Consequently, standard simplex pivots exhibit severe basis degeneracy: the solver may cycle, arbitrarily allocate consumption to younger stock, or produce non-FIFO fractional vertices without any objective penalty. By imposing strictly increasing holding costs $h_a = h + (a-1)\epsilon$ with $\epsilon = 0.01$, the reduced cost of leaving an older cohort unconsumed while depleting younger inventory is strictly positive (at least $\epsilon > 0$). This strictly breaks primal and dual simplex degeneracy, guaranteeing that non-FIFO bases cannot be optimal and establishing that the unique cost-minimizing LP basis and MILP vertex strictly satisfy FIFO physical discipline.

3.3.4 Post-Decision FIFO Evaluation Simulator

While Lemma 1 proves that the forward optimization model plans under strict FIFO consumption, real-world deployment requires a deterministic physical evaluation simulator to track inventory transitions and cost realizations when planned deliveries $(\mathbf{x}, \mathbf{q})$ encounter true out-of-sample customer demand $\tilde{d}_{i,t}$.

Algorithm 1 specifies the post-decision evaluation policy. It depletes inventory by age and, if post-demand carryover exceeds C_i, discards excess inventory from the oldest remaining cohort first. This policy enforces the store-capacity constraint in (11). The optimization model represents disposal with cohort-specific variables $\delta_{i,t,a}$; unless the same oldest-first rule is imposed there, the simulator and optimization model need not select identical disposal decisions. Accordingly, reported out-of-sample costs are conditional on the stated simulator policy and should not be interpreted as an exact match to optimized recourse without an implementation check.

Algorithm 1 Post-Decision FIFO Perishable Inventory Evaluation Simulator

Require: Planned delivery schedule $\mathbf{q} \in \mathbb{R}_+^{N\times T}$, routing plan $\mathbf{x} \in \{0,1\}^{(N+1)\times(N+1)\times K\times T}$, true demand realization $\tilde{\mathbf{d}} \in \mathbb{R}_+^{N\times T}$, shelf life τ, store capacities C_i, cost parameters (h_a, p_s, p_u, c_{ij}), initial stock $I_{i,0,a}$ for $a \in \{1,\dots,\tau-1\}$.

Ensure: Realized costs $(C_{\text{route}}, C_{\text{hold}}, C_{\text{spoil}}, C_{\text{stock}}, C_{\text{total}})$, carryover trajectories $\mathbf{I}$, consumption $\mathbf{w}$, recourse $(\mathbf{y}^+, \mathbf{y}^-)$.

$C_{\text{route}} \leftarrow \sum_{t=1}^{T}\sum_{k=1}^{K}\sum_{i=0}^{N}\sum_{j=0}^{N} c_{ij}x_{ijt}^k$
Initialize carryover buffer: $\mathbf{I}_{i,a}^{\text{prev}} \leftarrow I_{i,0,a}$ for all $i \in \{1,\dots,N\}, a \in \{1,\dots,\tau-1\}$
for $t = 1$ **to** T **do**
  for $i = 1$ **to** N **do**
    Assemble available stock vector $\mathbf{S} \in \mathbb{R}_+^{\tau}$:
    $S_1 \leftarrow q_{it}$ ▷ Fresh arrival at age 1
    for $a = 2$ **to** τ **do**
      $S_a \leftarrow \mathbf{I}_{i,a-1}^{\text{prev}}$ ▷ Previous stock aged by 1 period
    end for
    $d_{\text{rem}} \leftarrow \tilde{d}_{i,t}$
    Initialize consumption vector: $w_{i,t,a} \leftarrow 0$ for all $a \in \{1,\dots,\tau\}$ ▷ **FIFO Depletion: Oldest cohorts consumed first**
    for $a = \tau$ **down to** 1 **do**
      if $d_{\text{rem}} > 0$ **then**
        $w_{i,t,a} \leftarrow \min(S_a, d_{\text{rem}})$
        $d_{\text{rem}} \leftarrow d_{\text{rem}} - w_{i,t,a}$
        $S_a \leftarrow S_a - w_{i,t,a}$
      end if
    end for
    ▷ **Recourse and State Transitions**
    $y_{it}^+ \leftarrow \max(0, d_{\text{rem}})$ ▷ Unmet customer demand (stockout)
    $y_{it}^- \leftarrow \max(0, S_\tau)$ ▷ Expired age-τ inventory (spoilage)
    $E \leftarrow \max(0, \sum_{a=1}^{\tau-1} S_a - C_i)$
    for $a = \tau - 1$ **down to** 1 **do**
      $\delta_{i,t,a} \leftarrow \min(S_a, E)$ ▷ Discard oldest stock first if capacity is exceeded
      $S_a \leftarrow S_a - \delta_{i,t,a}$
      $E \leftarrow E - \delta_{i,t,a}$
      $y_{it}^- \leftarrow y_{it}^- + \delta_{i,t,a}$
    end for
    for $a = 1$ **to** $\tau - 1$ **do**
      $I_{i,t,a} \leftarrow S_a$ ▷ Carryover inventory of age a
      $\mathbf{I}_{i,a}^{\text{prev}} \leftarrow S_a$
    end for
  end for
end for
$C_{\text{hold}} \leftarrow \sum_{t=1}^{T}\sum_{i=1}^{N}\sum_{a=1}^{\tau-1} h_a I_{i,t,a}$
$C_{\text{spoil}} \leftarrow p_s \sum_{t=1}^{T}\sum_{i=1}^{N} y_{it}^-$
$C_{\text{stock}} \leftarrow p_u \sum_{t=1}^{T}\sum_{i=1}^{N} y_{it}^+$
$C_{\text{total}} \leftarrow C_{\text{route}} + C_{\text{hold}} + C_{\text{spoil}} + C_{\text{stock}}$
return $(C_{\text{route}}, C_{\text{hold}}, C_{\text{spoil}}, C_{\text{stock}}, C_{\text{total}}), \mathbf{I}, \mathbf{w}, \mathbf{y}^+, \mathbf{y}^-$

3.4 Fleet Routing, Operational Constraints, and Solver Execution Modes

The delivery fleet operates capacitated tours originating and terminating at the central depot. The complete vehicle routing and delivery formulation comprises:

$$\sum_{j\in\mathcal{N}} x^k_{0jt} \le 1, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \tag{14}$$

$$\sum_{j\in\mathcal{V},j\neq i} x^k_{jit} - \sum_{j\in\mathcal{V},j\neq i} x^k_{ijt} = 0, \quad \forall i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}, \tag{15}$$

$$\sum_{j\in\mathcal{N}} x^k_{0jt} - \sum_{i\in\mathcal{N}} x^k_{i0t} = 0, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \tag{16}$$

$$q^k_{it} \le \min(Q, C_i) \sum_{j\in\mathcal{V},j\neq i} x^k_{jit}, \quad \forall i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}, \tag{17}$$

$$\sum_{i\in\mathcal{N}} q^k_{it} \le Q \sum_{j\in\mathcal{N}} x^k_{0jt}, \quad \forall k \in \mathcal{K}, t \in \mathcal{T}, \tag{18}$$

$$\sum_{j\in\mathcal{V},j\neq i} x^k_{jit} \le \sum_{j\in\mathcal{N}} x^k_{0jt}, \quad \forall i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}, \tag{19}$$

$$\sum_{k\in\mathcal{K}} \sum_{j\in\mathcal{V},j\neq i} x^k_{jit} \le 1, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{20}$$

$$\sum_{j\in\mathcal{N}} x^k_{0jt} \le \sum_{j\in\mathcal{N}} x^{k-1}_{0jt}, \quad \forall k \in \{2,\dots,K\}, t \in \mathcal{T}, \tag{21}$$

$$u^k_{it} - u^k_{jt} + N x^k_{ijt} \le N - 1, \quad \forall i,j \in \mathcal{N}, i \neq j, k \in \mathcal{K}, t \in \mathcal{T}, \tag{22}$$

$$1 \le u^k_{it} \le N, \quad \forall i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}. \tag{23}$$

Constraints (14)–(16) enforce tour departure and return balance. Constraints (17)–(18) link delivered quantities to vehicle visits and enforce vehicle payload capacity Q. Constraint (19) ensures a vehicle cannot visit a store unless it is dispatched from the depot. Constraint (20) restricts each store to at most one vehicle visit per period, preventing fragmented partial drops. Constraint (21) breaks vehicle indexing symmetry by requiring vehicle k to be dispatched only if vehicle $k-1$ is active, eliminating $K!$ symmetric permutations in branch-and-cut exploration. Constraints (22)–(23) prevent disconnected sub-circuits via MTZ formulation [**?**].

Solver Execution Modes: The mathematical model is executed across two distinct solver modes:

(i) **Continuous LP Relaxation Mode**: Arc variables are relaxed to continuous bounds ($x^k_{ijt} \in [0,1]$). The resulting continuous LP is solved to dual optimality via SciPy/HiGHS with interior-point or dual-simplex algorithms, executing in ~ 0.40 seconds per epoch to provide smooth subgradient feedback during Phase 1 warmup.

(ii) **Discrete MILP Matheuristic Mode**: Solved via a cluster-first route-second matheuristic with localized branch-and-cut. This matheuristic clusters active store visits from the LP relaxation, constructs initial feasible tours using a greedy nearest-neighbor insertion with 2-opt edge-exchange local search, and solves a localized branch-and-cut MILP via HiGHS with dynamic subtour cut separation to obtain integer-optimal routing plans.

3.5 Two-Stage Recourse Decomposition and Bilevel Dual Subgradients

In conventional combinatorial formulations, uncertain customer demand $\tilde{d}_{i,t}$ appears on the right-hand side of inventory balance constraints (4). When developing Decision-Focused Learning (DFL) methodologies, this constraint structure presents a major mathematical hurdle: standard decision-focused surrogate losses (such as SPO+ [**?**] and IntOpt [**?**]) require optimization over

a fixed, deterministic decision space where predicted parameters reside exclusively in the linear objective function. If constraints depend directly on predicted demand $\hat{d}$, estimation errors can induce constraint infeasibility ($\mathcal{W}(\hat{d}) = \emptyset$), destroying the continuity of the downstream argmin operator and halting neural network gradient propagation.

To rigorously resolve this challenge without circular dependencies, we formulate the multi-period PIRP as a **Two-Stage Stochastic Program with Complete Linear Recourse**. We temporally and structurally decompose the decision space into:

(i) **First-Stage Tactical Decisions ($\mathbf{z}_1 \in \mathcal{W}_1$)**: Here-and-now vehicle dispatch, routing tours, visit sequencing, and fresh inventory delivery quantities committed at the beginning of each period prior to the realization of customer demand.

(ii) **Second-Stage Operational Recourse ($\mathbf{z}_2 \in \mathcal{W}_2(\mathbf{q}, \tilde{d})$)**: Wait-and-see FIFO stock consumption, emergency stockout fulfillment (y^+_{it}), spoilage disposal (y^-_{it}), and carryover inventory storage evaluated after customer demand $\tilde{d}$ unfolds on store shelves.

Definition 2 (First-Stage Feasible Decision Set $\mathcal{W}_1$)**.** Let $\mathbf{z}_1 = (\mathbf{x}, \mathbf{u}, \mathbf{q})$ denote the first-stage decision tuple comprising binary arc traversals $x^k_{ijt} \in \{0,1\}$, MTZ visit order auxiliaries $u^k_{it} \in [1, N]$, and delivery quantities $q^k_{it} \geq 0$ for all $(i,j) \in \mathcal{E}, k \in \mathcal{K}, t \in \mathcal{T}$. The deterministic first-stage feasible set $\mathcal{W}_1 \subset \{0,1\}^{|\mathcal{E}|KT} \times [1,N]^{NKT} \times \mathbb{R}^{NKT}_+$ is defined by constraints (14)–(23) alongside individual store delivery bounds:

$$\sum_{k \in \mathcal{K}} q^k_{it} \leq C_i, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}. \tag{24}$$

Notice that *every constraint defining $\mathcal{W}_1$ depends solely on fixed physical infrastructure parameters* (N, T, K, Q, C_i). The uncertain customer demand $\tilde{d}$ appears nowhere in the definition of $\mathcal{W}_1$.

Definition 3 (Second-Stage Recourse Feasible Set $\mathcal{W}_2(\mathbf{q}, \tilde{d})$)**.** Given aggregate replenishment deliveries $q_{it} = \sum_{k \in \mathcal{K}} q^k_{it}$ resulting from a first-stage decision $(\mathbf{x}, \mathbf{u}, \mathbf{q}) \in \mathcal{W}_1$ and any realized demand trajectory $\tilde{d} \in \mathbb{R}^{N \times T}_+$, the second-stage recourse decision tuple $\mathbf{z}_2 = (\mathbf{I}, \mathbf{w}, \mathbf{y}^+, \mathbf{y}^-, \boldsymbol{\delta})$ belongs to the recourse set $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ defined by:

$$\sum_{a=1}^{\tau} w_{i,t,a} + y^+_{it} = \tilde{d}_{i,t}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{25}$$

$$I_{i,t,1} + w_{i,t,1} + \delta_{i,t,1} = q_{it}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{26}$$

$$I_{i,t,a} + w_{i,t,a} + \delta_{i,t,a} = I_{i,t-1,a-1}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, a \in \{2, \ldots, \tau - 1\}, \tag{27}$$

$$w_{i,t,\tau} + \delta_{i,t,\tau} = I_{i,t-1,\tau-1}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{28}$$

$$y^-_{it} = \sum_{a=1}^{\tau} \delta_{i,t,a}, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{29}$$

$$\sum_{a=1}^{\tau-1} I_{i,t,a} \leq C_i, \quad \forall i \in \mathcal{N}, t \in \mathcal{T}, \tag{30}$$

$$I_{i,t,a} \geq 0, \quad w_{i,t,a} \geq 0, \quad \delta_{i,t,a} \geq 0, \quad y^+_{it} \geq 0, \quad y^-_{it} \geq 0, \quad \forall i, t, a, \tag{31}$$

where $I_{i,0,a} \geq 0$ represents known initial store inventory of age $a \in \{1, \ldots, \tau - 1\}$. The optimal second-stage recourse cost is given by:

$$\mathcal{Q}(\mathbf{q}, \tilde{d}) = \min_{(\mathbf{I}, \mathbf{w}, \mathbf{y}^+, \mathbf{y}^-, \boldsymbol{\delta}) \in \mathcal{W}_2(\mathbf{q}, \tilde{d})} \sum_{t=1}^{T} \sum_{i=1}^{N} \left(\sum_{a=1}^{\tau-1} h_a I_{i,t,a} + p_s y^-_{it} + p_u y^+_{it} \right). \tag{32}$$

Bilevel Recourse Formulation and True Task Loss: In the forward planning phase, given a predicted customer demand trajectory $\hat{\mathbf{d}} = f_\theta(\mathbf{X}) \in \mathbb{R}_+^{N\times T}$, the planner commits to the first-stage vehicle routing tours $\mathbf{x}^*(\hat{\mathbf{d}})$ and store replenishment delivery quantities $\mathbf{q}^*(\hat{\mathbf{d}})$ by solving the deterministic surrogate PIRP:

$$\mathbf{z}_1^*(\hat{\mathbf{d}}) = (\mathbf{x}^*(\hat{\mathbf{d}}), \mathbf{u}^*(\hat{\mathbf{d}}), \mathbf{q}^*(\hat{\mathbf{d}})) \in \arg\min_{(\mathbf{x},\mathbf{u},\mathbf{q})\in\mathcal{W}_1} \left\{ \mathbf{c}_{\text{route}}^T\mathbf{x} + \mathcal{Q}(\mathbf{q}, \hat{\mathbf{d}}) \right\}, \tag{33}$$

where $\mathbf{c}_{\text{route}}^T\mathbf{x} = \sum_{t=1}^T \sum_{k=1}^K \sum_{(i,j)\in\mathcal{E}} c_{ij} x_{ijt}^k$.

When the true customer demand trajectory $\mathbf{d} \in \mathbb{R}_+^{N\times T}$ is subsequently realized on store shelves, the first-stage delivery schedule $\mathbf{q}^*(\hat{\mathbf{d}})$ is already fixed. The actual operational cost incurred across the supply chain is governed by the **True Task Loss**:

$$\mathcal{L}_{\text{task}}(\hat{\mathbf{d}}; \mathbf{d}) = \mathbf{c}_{\text{route}}^T\mathbf{x}^*(\hat{\mathbf{d}}) + \mathcal{Q}(\mathbf{q}^*(\hat{\mathbf{d}}), \mathbf{d}), \tag{34}$$

where $\mathcal{Q}(\mathbf{q}^*(\hat{\mathbf{d}}), \mathbf{d})$ is the second-stage recourse linear program evaluated under the *ground-truth* demand $\mathbf{d}$. Subtracting the clairvoyant oracle cost $C^*(\mathbf{d}) = \min_{(\mathbf{x},\mathbf{u},\mathbf{q})\in\mathcal{W}_1}\{\mathbf{c}_{\text{route}}^T\mathbf{x} + \mathcal{Q}(\mathbf{q}, \mathbf{d})\}$ yields the exact Decision Regret $\mathcal{R}(\hat{\mathbf{d}}; \mathbf{d}) = \mathcal{L}_{\text{task}}(\hat{\mathbf{d}}; \mathbf{d}) - C^*(\mathbf{d}) \geq 0$.

Exact Recourse Subgradient via Linear Programming Duality: To train the neural network weights θ end-to-end to minimize $\mathcal{L}_{\text{task}}(\hat{\mathbf{d}}; \mathbf{d})$, the gradient chain rule requires differentiating the recourse function with respect to planned deliveries $\mathbf{q}$:

$$\nabla_\theta \mathcal{L}_{\text{task}}(\hat{\mathbf{d}}; \mathbf{d}) = \left(\frac{\partial \hat{\mathbf{d}}}{\partial \theta}\right)^T \left[\left(\frac{\partial \mathbf{x}^*(\hat{\mathbf{d}})}{\partial \hat{\mathbf{d}}}\right)^T \mathbf{c}_{\text{route}} + \left(\frac{\partial \mathbf{q}^*(\hat{\mathbf{d}})}{\partial \hat{\mathbf{d}}}\right)^T \nabla_{\mathbf{q}} \mathcal{Q}(\mathbf{q}^*(\hat{\mathbf{d}}), \mathbf{d}) \right]. \tag{35}$$

We now establish that $\nabla_{\mathbf{q}} \mathcal{Q}(\mathbf{q}^*, \mathbf{d})$ is obtained analytically and exactly from the dual multipliers of the second-stage recourse LP.

Proposition 1 (Exact Recourse Subgradient w.r.t. First-Stage Deliveries)**.** *For any feasible first-stage delivery schedule $\mathbf{q} \in \mathbb{R}_+^{N\times T}$ and realized demand trajectory $\mathbf{d} \in \mathbb{R}_+^{N\times T}$, the recourse function $\mathcal{Q}(\mathbf{q}, \mathbf{d})$ defined in (32) is a finite, continuous, convex, and piecewise-linear function of $\mathbf{q}$. By linear programming duality and Danskin's Theorem, a subgradient of $\mathcal{Q}(\mathbf{q}, \mathbf{d})$ with respect to the delivery quantity q_{it} is given directly by the optimal dual multiplier $\pi_{it}^*(\mathbf{q}, \mathbf{d})$ associated with the fresh-inventory balance constraint (26):*

$$\frac{\partial \mathcal{Q}(\mathbf{q}, \mathbf{d})}{\partial q_{it}} = \pi_{it}^*(\mathbf{q}, \mathbf{d}) \in [-p_u,\, p_s]\,. \tag{36}$$

Proof. For fixed realized demand $\mathbf{d}$ and fixed deliveries $\mathbf{q}$, $\mathcal{Q}(\mathbf{q}, \mathbf{d})$ is the optimal value of the linear program (32) over $\mathcal{W}_2(\mathbf{q}, \mathbf{d})$. In this LP, q_{it} appears exclusively on the right-hand side of constraint (26): $I_{i,t,1} + w_{i,t,1} + \delta_{i,t,1} = q_{it}$. By strong linear programming duality, $\mathcal{Q}(\mathbf{q}, \mathbf{d})$ equals the optimal objective value of its dual linear program. By Danskin's Theorem [**?**], the subdifferential of the optimal value of a linear program with respect to its right-hand-side vector is the set of optimal dual solutions. Hence, $\partial_{q_{it}} \mathcal{Q}(\mathbf{q}, \mathbf{d}) = \pi_{it}^*(\mathbf{q}, \mathbf{d})$.

The bounds $\pi_{it}^* \in [-p_u, p_s]$ follow from the extreme marginal cost realizations in the primal recourse problem: delivering one additional unit of fresh inventory can at best avoid an emergency stockout penalty p_u (yielding marginal cost $\pi_{it}^* \geq -p_u$), and at worst incur disposal plus cumulative holding costs p_s (yielding $\pi_{it}^* \leq p_s$). Boundedness and convexity follow from the general properties of two-stage linear recourse functions. □

Economic Mechanism and First-Stage Linearized Surrogate: Proposition 1 reveals an essential economic mechanism: the optimal dual multiplier $\pi^*_{it}(\mathbf{q}^*(\hat{\mathbf{d}}), \mathbf{d})$ directly encodes the asymmetric operational penalties without any ad-hoc scaling. If store i experiences unmet customer demand during period t (or subsequent periods before shelf-life expiration), an additional unit of delivery q_{it} eliminates emergency fulfillment, so $\pi^*_{it} \approx -p_u < 0$. Conversely, if store i holds excess inventory that spoils unconsumed at age τ, an additional unit of delivery escalates holding and disposal costs, so $\pi^*_{it} > 0$.

Linearizing the second-stage recourse cost around planned deliveries $\mathbf{q}^*(\hat{\mathbf{d}})$ yields the first-stage linearized surrogate loss on $\mathcal{W}_1$:

$$\mathcal{L}_{\text{lin}}(\mathbf{x}, \mathbf{q}; \mathbf{d}) = \sum_{t=1}^{T} \sum_{k=1}^{K} \sum_{(i,j)\in\mathcal{E}} c_{ij} x^k_{ijt} + \sum_{t=1}^{T} \sum_{i=1}^{N} \pi^*_{it}(\mathbf{q}^*(\hat{\mathbf{d}}), \mathbf{d}) q_{it}. \tag{37}$$

Because the first-stage constraint set $\mathcal{W}_1$ is completely independent of customer demand (Definition 2), $\mathcal{L}_{\text{lin}}(\mathbf{x}, \mathbf{q}; \mathbf{d})$ provides a mathematically rigorous surrogate objective for downstream DFL backpropagation.

3.6 Theoretical Guarantees: Complete Second-Stage Recourse and Compactness

We now state and prove the foundational theorem establishing that the two-stage decision-focused optimization layer is mathematically well-posed and universally immune to solver infeasibility.

Theorem 1 (Complete Second-Stage Recourse and Compactness)**.** *Let $\mathcal{W}_1$ be the first-stage feasible set defined by constraints (14)–(24), and let $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ be the second-stage recourse set defined by (25)–(31). Then:*

(i) ***Demand-Independence and Compactness****: The first-stage feasible set $\mathcal{W}_1$ is strictly independent of customer demand $\tilde{d}$. Furthermore, $\mathcal{W}_1$ is non-empty ($\mathcal{W}_1 \neq \emptyset$) and topologically compact in Euclidean space.*

(ii) ***Complete Recourse with Retail Storage Capacity****: For every first-stage decision $(\mathbf{x}, \mathbf{u}, \mathbf{q}) \in \mathcal{W}_1$ and every non-negative customer demand realization $\tilde{d} \in \mathbb{R}^{N\times T}_+$, the second-stage recourse set $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ is non-empty:*

$$\forall (\mathbf{x}, \mathbf{u}, \mathbf{q}) \in \mathcal{W}_1, \quad \forall \tilde{d} \in \mathbb{R}^{N\times T}_+ \implies \mathcal{W}_2(\mathbf{q}, \tilde{d}) \neq \emptyset.$$

This holds unconditionally even under zero customer demand ($\tilde{d}_{i,t} = 0$) when deliveries and entering carryover exceed retail capacity ($I_{i,t-1} + q_{it} > C_i$).

(iii) ***Finite Recourse Value****: For strictly non-negative unit cost parameters $h_a \geq 0, p_s \geq 0, p_u \geq 0$, the optimal recourse cost $\mathcal{Q}(\mathbf{q}, \tilde{d})$ is non-negative and strictly finite:*

$$0 \leq \mathcal{Q}(\mathbf{q}, \tilde{d}) \leq \sum_{t=1}^{T} \sum_{i=1}^{N} \left[p_u \tilde{d}_{i,t} + (h_{\tau-1}\tau + p_s) q_{it} \right] + \sum_{i=1}^{N} (h_{\tau-1}\tau + p_s) \sum_{a=1}^{\tau-1} I_{i,0,a} < +\infty.$$

(iv) ***Universal Solver Safety****: For any predicted demand vector $\hat{d} \in \mathbb{R}^{N\times T}_+$ output by a machine learning model, the downstream optimization problem:*

$$\mathbf{z}^*_1(\hat{d}) \in \arg\min_{(\mathbf{x},\mathbf{u},\mathbf{q})\in\mathcal{W}_1} \left\{ \sum_{t=1}^{T} \sum_{k=1}^{K} \sum_{(i,j)\in\mathcal{E}} c_{ij} x^k_{ijt} + \mathcal{Q}(\mathbf{q}, \hat{d}) \right\}$$

possesses at least one optimal solution, and the solver never terminates with an infeasible status.

Proof. We prove each statement sequentially:

Part 1. Demand-Independence and Compactness of $\mathcal{W}_1$: Inspection of constraints (14)–(24) confirms that no constraint contains the demand vector $\tilde{d}$. Thus $\mathcal{W}_1$ is demand-independent.

To prove non-emptiness ($\mathcal{W}_1 \neq \emptyset$), consider the candidate null dispatch decision $\mathbf{z}_1^{(0)} = (\mathbf{x}^{(0)}, \mathbf{u}^{(0)}, \mathbf{q}^{(0)})$:

- Set $x_{ijt}^{k,(0)} = 0$ for all $(i,j) \in \mathcal{E}, k \in \mathcal{K}, t \in \mathcal{T}$;
- Set $u_{it}^{k,(0)} = 1$ for all $i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}$;
- Set $q_{it}^{k,(0)} = 0$ for all $i \in \mathcal{N}, k \in \mathcal{K}, t \in \mathcal{T}$.

We verify constraint satisfaction for $\mathbf{z}_1^{(0)}$: Depot departure (14) yields $0 \leq 1$. Degree and flow balance (15)–(16) yield $0 - 0 = 0$. Delivery linking (17) yields $0 \leq \min(Q, C_i) \cdot 0 = 0$. Vehicle capacity (18) yields $0 \leq Q \cdot 0 = 0$. Visit linking (19) yields $0 \leq 0$. Single-visit (20) yields $0 \leq 1$. Symmetry breaking (21) yields $0 \leq 0$. MTZ constraints (22) become $1 - 1 + N \cdot 0 = 0 \leq N - 1$, which holds strictly since $N \geq 2$. Store capacity (24) yields $0 \leq C_i$. All integrality and continuous bound constraints $x_{ijt}^k \in \{0,1\}$, $u_{it}^k \in [1, N]$, $q_{it}^k \geq 0$ are satisfied. Thus $\mathbf{z}_1^{(0)} \in \mathcal{W}_1$, proving $\mathcal{W}_1 \neq \emptyset$.

To prove compactness, observe that $\mathcal{W}_1$ is the intersection of hyperplanes and closed half-spaces in Euclidean space with the discrete integer set $\{0,1\}^{|\mathcal{E}|KT} \times [1, N]^{NKT} \times [0, \min(Q, C_i)]^{NKT}$. The set $\{0,1\}$ is closed in $\mathbb{R}$. As the finite intersection of closed sets, $\mathcal{W}_1$ is closed. Furthermore, every coordinate of $\mathbf{z}_1$ is bounded: $x_{ijt}^k \in [0,1]$, $u_{it}^k \in [1, N]$, and $q_{it}^k \in [0, \min(Q, C_i)]$. By the Heine-Borel theorem, every closed and bounded subset of Euclidean space is compact. Therefore, $\mathcal{W}_1$ is compact.

Part 2. Complete Second-Stage Recourse with Retail Storage Capacity ($\mathcal{W}_2(\mathbf{q}, \tilde{d}) \neq \emptyset$): Let $(\mathbf{x}, \mathbf{u}, \mathbf{q}) \in \mathcal{W}_1$ be an arbitrary first-stage plan, and let $\tilde{d} \in \mathbb{R}_+^{N \times T}$ be an arbitrary non-negative demand realization. We explicitly construct a feasible second-stage recourse tuple $\mathbf{z}_2 = (\mathbf{I}, \mathbf{w}, \mathbf{y}^+, \mathbf{y}^-, \boldsymbol{\delta})$ by forward induction over periods $t = 1, 2, \ldots, T$.

At period t, for each retail store $i \in \mathcal{N}$, available inventory entering period t by age category is:

$$S_{i,t,1} = q_{it} \geq 0, \quad S_{i,t,a} = I_{i,t-1,a-1} \geq 0 \quad (\forall a \in \{2, \ldots, \tau\}),$$

where $I_{i,0,a-1} \geq 0$ is the known initial stock. Total available stock at store i is $S_{it} = \sum_{a=1}^{\tau} S_{i,t,a} \geq 0$. We allocate consumption greedily under FIFO order, consuming oldest stock first ($a = \tau, \tau - 1, \ldots, 1$). Let $R_{i,t,\tau} = \tilde{d}_{i,t} \geq 0$ denote initial demand to be fulfilled. For $a = \tau, \tau - 1, \ldots, 1$, define:

$$w_{i,t,a} = \min\left(S_{i,t,a}, R_{i,t,a}\right) \geq 0,$$
$$R_{i,t,a-1} = R_{i,t,a} - w_{i,t,a} \geq 0.$$

Stockout recourse is set to:

$$y_{it}^+ = R_{i,t,0} = \max\left(0, \tilde{d}_{i,t} - S_{it}\right) \geq 0,$$

which guarantees $\sum_{a=1}^{\tau} w_{i,t,a} + y_{it}^+ = \tilde{d}_{i,t}$, satisfying (25).

For goods reaching expiration age τ, all remaining stock spoils naturally:

$$\delta_{i,t,\tau} = S_{i,t,\tau} - w_{i,t,\tau} = I_{i,t-1,\tau-1} - w_{i,t,\tau} \geq 0,$$

satisfying (28).

For intermediate cohorts $a \in \{1, \ldots, \tau - 1\}$, tentative carryover stock before excess disposal is:

$$\hat{I}_{i,t,a} = S_{i,t,a} - w_{i,t,a} \geq 0.$$

Total tentative carryover is $\hat{I}_{it} = \sum_{a=1}^{\tau-1} \hat{I}_{i,t,a} \geq 0$. We consider two exhaustive cases:

- *Case A ($\hat{I}_{it} \le C_i$)*: Physical retail storage capacity is not exceeded. Set proactive disposal slacks $\delta_{i,t,a} = 0$ and final carryovers $I_{i,t,a} = \hat{I}_{i,t,a} \ge 0$ for all $a \in \{1, \dots, \tau - 1\}$.
- *Case B ($\hat{I}_{it} > C_i$)*: Total surviving inventory exceeds store capacity C_i (e.g., when $I_{i,t-1} + q_{it} > C_i$ and demand $\tilde{d}_{i,t}$ is near zero). Define excess stock $E_{it} = \hat{I}_{it} - C_i > 0$. We allocate proactive disposal slacks $\delta_{i,t,a} \ge 0$ greedily from the oldest intermediate cohorts ($a = \tau - 1$ down to 1) such that:

$$\sum_{a=1}^{\tau-1} \delta_{i,t,a} = E_{it}, \quad \text{with } 0 \le \delta_{i,t,a} \le \hat{I}_{i,t,a}.$$

  Set final carryovers $I_{i,t,a} = \hat{I}_{i,t,a} - \delta_{i,t,a} \ge 0$. Then total carryover satisfies:

$$\sum_{a=1}^{\tau-1} I_{i,t,a} = \sum_{a=1}^{\tau-1} (\hat{I}_{i,t,a} - \delta_{i,t,a}) = \hat{I}_{it} - E_{it} = C_i \le C_i,$$

  strictly satisfying retail storage capacity (30).

Total discarded inventory is set to $y^-_{it} = \sum_{a=1}^{\tau} \delta_{i,t,a} \ge 0$, satisfying (29). Balance equations (26) and (27) hold by construction ($I_{i,t,a} + w_{i,t,a} + \delta_{i,t,a} = \hat{I}_{i,t,a} + w_{i,t,a} = S_{i,t,a}$). All variables are non-negative. Hence, $(\mathbf{I}, \mathbf{w}, \mathbf{y}^+, \mathbf{y}^-, \boldsymbol{\delta}) \in \mathcal{W}_2(\mathbf{q}, \tilde{d})$, proving that $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ is non-empty for every $(\mathbf{x}, \mathbf{u}, \mathbf{q}) \in \mathcal{W}_1$ and every $\tilde{d} \ge 0$.

Part 3. Finite Recourse Value: Because $h_a \ge 0, p_s \ge 0, p_u \ge 0$ and all variables in $\mathcal{W}_2(\mathbf{q}, \tilde{d})$ are non-negative, $\mathcal{Q}(\mathbf{q}, \tilde{d}) \ge 0$. To establish the upper bound, evaluate the candidate solution constructed in Part 2: Stockouts satisfy $y^+_{it} \le \tilde{d}_{i,t}$. Because items expire after τ periods and retail storage capacity is capped at C_i, any delivered quantity q_{it} and initial stock $I_{i,0,a}$ can remain in inventory for at most τ periods before either being consumed, spoiled, or proactively discarded. Therefore, $\sum_{t=1}^{T} \sum_{a=1}^{\tau-1} I_{i,t,a} \le \tau \left(\sum_{t=1}^{T} q_{it} + \sum_{a=1}^{\tau-1} I_{i,0,a} \right)$, and cumulative disposal satisfies $\sum_{t=1}^{T} y^-_{it} \le \sum_{t=1}^{T} q_{it} + \sum_{a=1}^{\tau-1} I_{i,0,a}$. Substituting these bounds into (32) gives:

$$\mathcal{Q}(\mathbf{q}, \tilde{d}) \le \sum_{t=1}^{T} \sum_{i=1}^{N} p_u \tilde{d}_{i,t} + \sum_{i=1}^{N} (h_{\tau-1}\tau + p_s) \left(\sum_{t=1}^{T} q_{it} + \sum_{a=1}^{\tau-1} I_{i,0,a} \right).$$

Since $q_{it} \le C_i < \infty$, $I_{i,0,a} \le C_i < \infty$, and $\tilde{d}_{i,t} < \infty$, the upper bound is strictly finite, proving $0 \le \mathcal{Q}(\mathbf{q}, \tilde{d}) < +\infty$.

Part 4. Universal Solver Safety: For any predicted demand $\hat{d} \in \mathbb{R}_+^{N \times T}$, the recourse function $\mathcal{Q}(\mathbf{q}, \hat{d})$ is the optimal value of a feasible linear program with bounded objective coefficients. By linear programming duality, $\mathcal{Q}(\mathbf{q}, \hat{d})$ is a continuous, piecewise linear convex function of $\mathbf{q}$ on $\mathcal{W}_1$. Because transportation routing costs are linear and $\mathcal{W}_1$ is non-empty and compact (Part 1), the total objective function is continuous on a compact domain. By the Weierstrass Extreme Value Theorem, the objective achieves its global minimum on $\mathcal{W}_1$. Therefore, an optimal decision $\mathbf{z}_1^*(\hat{d})$ exists and the optimization solver is guaranteed never to fail due to infeasibility.

This completes the proof. □

Corollary 1 (Universal Solver Safety in DFL Pipelines)**.** *Because the first-stage feasibility set $\mathcal{W}_1$ is completely independent of $\hat{d}$ and the recourse function $\mathcal{Q}(\mathbf{q}, \hat{d})$ is finite everywhere on $\mathbb{R}_+^{N \times T}$, the combinatorial optimization layer acts as a well-defined mapping $\mathcal{S} : \mathbb{R}_+^{N \times T} \to \mathcal{W}_1$. No output generated by a neural network during training or testing can ever cause the optimization solver to encounter an infeasible problem.*

4 End-to-End Decision-Focused Learning Methodology

4.1 The Predict-then-Optimize Dilemma: Decision Regret

Let $x \in \mathbb{R}^p$ denote contextual feature vectors (historical lag sales, promotional banners, holiday indicators, local weather). A neural network $f_\theta : \mathbb{R}^p \to \mathbb{R}^{\dim(w)}$ parameterized by weights θ outputs a predicted cost vector $\hat{c} = f_\theta(x)$. The operator computes decisions:

$$w^*(\hat{c}) \in \arg\min_{w \in \mathcal{W}} \hat{c}^T w.$$

However, the true operational cost incurred in the supply chain is evaluated under the realized ground-truth cost vector c:

$$\text{Cost}(w^*(\hat{c}); c) = c^T w^*(\hat{c}).$$

The true operational loss is the **Decision Regret** (suboptimality gap):

$$\mathcal{R}(\hat{c}, c) = c^T w^*(\hat{c}) - c^T w^*(c) \geq 0, \tag{38}$$

where $w^*(c) \in \arg\min_{w \in \mathcal{W}} c^T w$ is the clairvoyant optimal decision with perfect information.

The Non-Differentiability Bottleneck: Because $\mathcal{W}$ contains discrete integer routing constraints, the solution operator $\hat{c} \mapsto w^*(\hat{c})$ is a piecewise constant step function:

$$\nabla_{\hat{c}} w^*(\hat{c}) = \mathbf{0} \quad \text{almost everywhere (a.e.).}$$

Consequently, applying the chain rule directly to Decision Regret yields:

$$\nabla_\theta \mathcal{R} = \left(\frac{\partial \hat{c}}{\partial \theta}\right)^T \cdot \nabla_{\hat{c}} w^*(\hat{c}) \cdot c = \mathbf{0} \quad \text{a.e.,}$$

providing zero directional signal for gradient descent. To train θ end-to-end, surrogate loss functions are required.

4.2 Surrogate Formulation 1: Smart Predict-then-Optimize (SPO+)

The SPO+ loss function [**?**] provides a convex surrogate upper bound to the non-convex decision regret.

Definition 4 (SPO+ Loss)**.** For a predicted cost vector $\hat{c}$ and true cost vector c, the SPO+ loss is defined as:

$$\ell_{\text{SPO+}}(\hat{c}, c) = \max_{w \in \mathcal{W}} \left\{ (c - 2\hat{c})^T w \right\} + 2\hat{c}^T w^*(c) - c^T w^*(c). \tag{39}$$

Theorem 2 (Convexity and Upper Bound Property of SPO+)**.** *For any true cost $c \in \mathbb{R}^{\dim(w)}$ and predicted cost $\hat{c} \in \mathbb{R}^{\dim(w)}$:*

(i) $\ell_{SPO+}(\hat{c}, c)$ is a convex function of $\hat{c}$.

(ii) $\ell_{SPO+}(\hat{c}, c) \geq \mathcal{R}(\hat{c}, c) \geq 0$.

Proof. We prove both parts:

Part 1. Convexity: In (39), the term $2\hat{c}^T w^*(c)$ is linear in $\hat{c}$, and $-c^T w^*(c)$ is constant with respect to $\hat{c}$. The first term is:

$$g(\hat{c}) = \max_{w \in \mathcal{W}} \left\{ (c - 2\hat{c})^T w \right\} = \max_{w \in \mathcal{W}} \left\{ c^T w - 2w^T \hat{c} \right\}.$$

For every fixed $w \in \mathcal{W}$, the function $\phi_w(\hat{c}) = c^T w - 2w^T \hat{c}$ is an affine function of $\hat{c}$. The pointwise supremum of any collection of affine functions is convex [**?**]. Because $\mathcal{W}$ is non-empty (Theorem 1), $g(\hat{c})$ is the pointwise maximum of affine functions and is therefore convex. The sum of a convex function and an affine function is convex. Thus, $\ell_{\text{SPO+}}(\hat{c}, c)$ is convex in $\hat{c}$.

Part 2. Upper Bound: By definition of the maximum over $\mathcal{W}$, for any feasible vector $\bar{w} \in \mathcal{W}$:

$$\max_{w \in \mathcal{W}} \left\{ (c - 2\hat{c})^T w \right\} \geq (c - 2\hat{c})^T \bar{w}.$$

Selecting $\bar{w} = w^*(\hat{c}) \in \mathcal{W}$ yields:

$$\max_{w \in \mathcal{W}} \left\{ (c - 2\hat{c})^T w \right\} \geq (c - 2\hat{c})^T w^*(\hat{c}) = c^T w^*(\hat{c}) - 2\hat{c}^T w^*(\hat{c}).$$

Substituting this lower bound into (39):

$$\begin{aligned} \ell_{\text{SPO+}}(\hat{c}, c) &\geq c^T w^*(\hat{c}) - 2\hat{c}^T w^*(\hat{c}) + 2\hat{c}^T w^*(c) - c^T w^*(c) \\ &= \left(c^T w^*(\hat{c}) - c^T w^*(c) \right) + 2\hat{c}^T \left(w^*(c) - w^*(\hat{c}) \right). \end{aligned} \tag{40}$$

By the optimality of $w^*(\hat{c})$ with respect to cost vector $\hat{c}$, we have $\hat{c}^T w^*(\hat{c}) \leq \hat{c}^T w$ for all $w \in \mathcal{W}$. In particular, for $w = w^*(c) \in \mathcal{W}$:

$$\hat{c}^T w^*(\hat{c}) \leq \hat{c}^T w^*(c) \iff \hat{c}^T \left(w^*(c) - w^*(\hat{c}) \right) \geq 0.$$

Multiplying by 2 preserves non-negativity: $2\hat{c}^T (w^*(c) - w^*(\hat{c})) \geq 0$. Substituting back into (40):

$$\ell_{\text{SPO+}}(\hat{c}, c) \geq c^T w^*(\hat{c}) - c^T w^*(c) = \mathcal{R}(\hat{c}, c) \geq 0.$$

This completes the proof. □

Contrastive RHS Demand Surrogate: Applying Danskin's Theorem to the classical SPO+ convex upper bound (39), the objective-space subgradient with respect to cost vector $\hat{c}$ on $\mathcal{W}_1$ is:

$$g_{\hat{c}} \in \partial_{\hat{c}} \ell_{\text{SPO+}}(\hat{c}, c) = 2 \left(w^*(c) - w^*(2\hat{c} - c) \right). \tag{41}$$

In inventory routing, however, uncertain customer demand $\hat{\mathbf{d}}$ resides on the *right-hand side* of inventory balance constraints rather than in the objective cost function. Furthermore, the true multi-period PIRP involves binary routing variables $\mathbf{x} \in \{0, 1\}$. The mapping $\hat{\mathbf{d}} \mapsto (\mathbf{x}^*(\hat{\mathbf{d}}), \mathbf{q}^*(\hat{\mathbf{d}}))$ is piecewise constant with zero derivatives almost everywhere and discrete jump discontinuities at polyhedral facet transitions. Consequently, true classical calculus gradients through the combinatorial MILP do not exist in standard analysis.

In Section 3.5, Proposition 1 establishes a recourse subgradient with respect to continuous replenishment decisions under the stated LP assumptions. This does not by itself give a gradient through predicted demand or the discrete routing solution. We therefore define the stop-gradient estimator below as a custom, contrastive demand-space update inspired by the SPO+ principle; we do not identify it with the classical SPO+ loss or claim it is an exact subgradient of realized MILP cost.

Definition 5 (Stop-Gradient RHS Demand-Space Surrogate)**.** Given predicted customer demand $\hat{\mathbf{d}} \in \mathbb{R}_+^{N \times T}$ and true demand $\mathbf{d} \in \mathbb{R}_+^{N \times T}$, define the contrastive demand request $\hat{\mathbf{d}}_{\text{spo}} = \max(\mathbf{0}, 2\hat{\mathbf{d}} - \mathbf{d})$. Let $\mathbf{q}^*(\hat{\mathbf{d}}_{\text{spo}})$ and $\mathbf{q}^*(\mathbf{d})$ denote the replenishment delivery allocations obtained by solving the forward surrogate optimization problem (33) evaluated at $\hat{\mathbf{d}}_{\text{spo}}$ and $\mathbf{d}$, respectively.

Let $\Delta\mathbf{q} = \mathbf{q}^*(\hat{\mathbf{d}}_{\text{spo}}) - \mathbf{q}^*(\mathbf{d}) \in \mathbb{R}^{N \times T}$ denote the contrastive replenishment shift. Define the dimensionless asymmetric weighting $\boldsymbol{\omega}(\Delta\mathbf{q}) \in \mathbb{R}_+^{N \times T}$:

$$\omega_{it}(\Delta q_{it}) = \begin{cases} \dfrac{p_u}{c_{\text{norm}}} & \text{if } \Delta q_{it} < 0 \quad \text{(under-delivery penalty)}, \\ \dfrac{c_o}{c_{\text{norm}}} & \text{if } \Delta q_{it} \geq 0 \quad \text{(over-delivery penalty)}, \end{cases} \tag{42}$$

where $c_o = h + \frac{p_s}{\tau}$ is the effective overage cost per unit per period, and $c_{\text{norm}} = \frac{p_u + c_o}{2}$ is the normalizing reference cost.

For any quantity z that depends on $\hat{\mathbf{d}}$, let $\text{sg}(z)$ denote a stop-gradient operator: it has forward value z and derivative zero with respect to $\hat{\mathbf{d}}$. The training surrogate is:

$$\begin{aligned}\mathcal{L}_{\text{SG}}(\hat{\mathbf{d}}; \mathbf{d}) = (2\hat{\mathbf{d}} - \mathbf{d})^T \,\text{sg}[\boldsymbol{\omega}(\Delta\mathbf{q}) \odot \Delta\mathbf{q}] \\ + \text{sg}\Big[\max\Big(0, \mathcal{C}(\mathbf{q}^*(\hat{\mathbf{d}}_{\text{spo}})) - \mathcal{C}(\mathbf{q}^*(\mathbf{d}))\Big)\Big]\,,\end{aligned} \tag{43}$$

where $\mathcal{C}(\mathbf{q}) = \mathbf{c}_{\text{route}}^T \mathbf{x}^* + \mathcal{Q}(\mathbf{q}, \mathbf{d})$ denotes total realized operational cost. All optimizer outputs, weights, and the cost-gap term are treated as constants during backpropagation. Thus, this construction defines a custom surrogate update; it is not the derivative or subgradient of the corresponding stop-gradient-free expression, nor is it claimed to be the classical SPO+ loss for RHS uncertainty.

Proposition 2 (Gradient of the Stop-Gradient Surrogate)**.** *With the stop-gradient convention in Definition 5, differentiation of $\mathcal{L}_{SG}$ with respect to $\hat{\mathbf{d}}$ gives:*

$$\mathbf{g}_{\hat{\mathbf{d}}}^{SG} = 2\,\text{sg}[\boldsymbol{\omega}(\Delta\mathbf{q}) \odot \Delta\mathbf{q}]\,. \tag{44}$$

This is the exact autodifferentiation gradient of the explicitly defined stop-gradient computational graph. It is a heuristic update direction for the decision problem, not a subgradient of the underlying MILP value function. In particular, shared vehicle and storage capacities couple store-period decisions, so the sign of an individual component is not guaranteed solely by whether $\hat{d}_{it}$ is above or below d_{it}.

(i) The weights and contrastive allocations are evaluated at the current predictions, then held fixed for that backward pass.

(ii) The stop-gradient construction avoids claiming differentiability of the discontinuous MILP solution map; it does not establish convexity or a global descent guarantee for realized operational cost.

Proof. By definition, the stop-gradient terms have zero derivative with respect to $\hat{\mathbf{d}}$. The derivative of the remaining linear term is therefore $2\,\text{sg}[\boldsymbol{\omega}(\Delta\mathbf{q}) \odot \Delta\mathbf{q}]$, proving (44). □

Equation (44) specifies a cost-weighted heuristic update based on contrastive replenishment decisions. Its behavior must be assessed empirically; the sign and magnitude of each component are not guaranteed to correspond to a local change in realized cost. The following sections describe separate smoothing approaches based on RHS interior-point differentiation and perturbed optimization.

4.3 Surrogate Formulation 2: RHS Interior Point Differentiation (RHS-IntOpt)

Following **?**], we differentiate linear relaxations using a logarithmic barrier function. Consider the continuous LP relaxation over polyhedron $\mathcal{P} = \{\mathbf{w} \geq \mathbf{0} : A\mathbf{w} = \mathbf{b}(\hat{\mathbf{d}})\}$, where $\mathbf{b}(\hat{\mathbf{d}}) = \mathbf{b}_0 + E_d\hat{\mathbf{d}}$, and E_d places predicted customer demand $\hat{\mathbf{d}}$ onto the right-hand side of demand constraints (25). Introducing barrier parameter $\mu > 0$:

$$\min_{\mathbf{w} \geq \mathbf{0}} \mathbf{c}^T\mathbf{w} - \mu \sum_{j=1}^{m} \ln(w_j) \quad \text{s.t. } A\mathbf{w} = \mathbf{b}_0 + E_d\hat{\mathbf{d}}. \tag{45}$$

Letting $\boldsymbol{\nu} \in \mathbb{R}^p$ denote equality multipliers and $\boldsymbol{\lambda} \in \mathbb{R}_+^m$ denote dual barrier multipliers, the perturbed KKT system is:

$$F(\mathbf{w}, \boldsymbol{\nu}, \boldsymbol{\lambda}; \hat{\mathbf{d}}) = \begin{bmatrix} \mathbf{c} - A^T\boldsymbol{\nu} - \boldsymbol{\lambda} \\ W\boldsymbol{\lambda} - \mu\mathbf{1} \\ A\mathbf{w} - \mathbf{b}_0 - E_d\hat{\mathbf{d}} \end{bmatrix} = \mathbf{0},$$

where $W = \mathrm{diag}(\mathbf{w})$. Eliminating $\mathrm{d}\boldsymbol{\lambda} = -\mu W^{-2}\mathrm{d}\mathbf{w}$ and applying the Implicit Function Theorem with respect to predicted demand $\hat{\mathbf{d}}$ yields the exact **RHS Sensitivity Jacobian**:

$$\frac{\partial \mathbf{w}^*(\hat{\mathbf{d}};\mu)}{\partial \hat{\mathbf{d}}} = W^2 A^T \left(AW^2A^T\right)^{-1} E_d. \tag{46}$$

Because AW^2A^T is symmetric positive definite (and already factorized at convergence of the interior-point solver), the RHS Jacobian is well-conditioned and stable. The gradient of the task loss with respect to network parameters θ is then:

$$\nabla_\theta \mathcal{L}_{\text{IntOpt}} = \left(\frac{\partial \hat{\mathbf{d}}}{\partial \theta}\right)^T \left(\frac{\partial \mathbf{w}^*(\hat{\mathbf{d}};\mu)}{\partial \hat{\mathbf{d}}}\right)^T \mathbf{c}_{\text{task}}. \tag{47}$$

4.4 Surrogate Formulation 3: RHS Differentiable Perturbed Optimizer (RHS-DPO)

Following **?**], combinatorial optimizers can also be smoothed by injecting random perturbations. In our right-hand-side setting, however, injecting unrestricted Gaussian noise $\hat{\mathbf{d}} + \sigma\mathbf{Z}$ directly into predicted demand introduces a critical domain violation: unbounded Gaussian noise produces negative demand realizations with non-zero probability ($\mathbb{P}(\hat{d}_{it} + \sigma Z_{it} < 0) > 0$). Negative customer demands violate the physical non-negativity constraint ($d_{it} \geq 0$) required by the PIRP formulation, generating ill-posed negative replenishment requests and invalidating the recourse feasibility guarantee of Theorem 1.

To guarantee strict operational validity while retaining the smoothing properties of perturbed optimizers, we introduce a **Projected Non-Negative Perturbation Scheme**:

$$\mathbf{d}_\sigma^{(m)} = \Pi_{\mathbb{R}_+^{N\times T}}\left(\hat{\mathbf{d}} + \sigma\mathbf{Z}^{(m)}\right) = \max\left(\mathbf{0}, \hat{\mathbf{d}} + \sigma\mathbf{Z}^{(m)}\right), \quad \mathbf{Z}^{(m)} \sim \mathcal{N}(\mathbf{0}, I), \tag{48}$$

where $\Pi_{\mathbb{R}_+^{N\times T}}(\cdot)$ denotes the Euclidean metric projection onto the non-negative orthant $\mathbb{R}_+^{N\times T}$, and $\sigma > 0$ is the perturbation bandwidth. The smoothed expected replenishment decision is estimated across M Monte Carlo perturbations:

$$\bar{\mathbf{q}}_\sigma^*(\hat{\mathbf{d}}) = \frac{1}{M}\sum_{m=1}^{M} \mathbf{q}^*\left(\mathbf{d}_\sigma^{(m)}\right). \tag{49}$$

The demand-space gradient of the smoothed task loss is then evaluated as:

$$\mathbf{g}_{\hat{\mathbf{d}}}^{\text{DPO}} = \boldsymbol{\omega}(\Delta\bar{\mathbf{q}}) \odot \left(\bar{\mathbf{q}}_\sigma^*(\hat{\mathbf{d}}) - \mathbf{q}^*(\mathbf{d})\right), \tag{50}$$

where $\Delta\bar{\mathbf{q}} = \bar{\mathbf{q}}_\sigma^*(\hat{\mathbf{d}}) - \mathbf{q}^*(\mathbf{d})$, and $\boldsymbol{\omega}$ applies the asymmetric Newsvendor weighting (42).

Analysis of the Projected Perturbation Estimator: The projected estimator (48)–(50) satisfies three key structural properties:

(i) **Guaranteed Solver Feasibility**: Because $\mathbf{d}_\sigma^{(m)} \in \mathbb{R}_+^{N\times T}$ for all $m \in \{1, \ldots, M\}$ by construction, every sampled optimization instance is strictly non-negative. By Theorem 1, complete recourse feasibility holds identically for every perturbed sample without numerical breakdown or solver infeasibility.

(ii) **Interior Truncation Probability**: For any store-period coordinate where predicted demand is strictly positive, the probability that the unprojected Gaussian perturbation falls into the negative truncation region is governed by the standard normal cumulative distribution function Φ:

$$\mathbb{P}(\hat{d}_{it} + \sigma Z_{it} < 0) = \Phi\left(-\frac{\hat{d}_{it}}{\sigma}\right). \tag{51}$$

In our calibrated grocery distribution network, daily demand averages $\bar{d} \approx 10$ to 25 units per store, while the perturbation bandwidth is set to $\sigma = 2.0$. For typical store demands ($\hat{d}_{it} \geq 10$), the normalized margin is $\frac{\hat{d}_{it}}{\sigma} \geq 5.0$, yielding an infinitesimal truncation probability $\Phi(-5.0) \approx 2.87 \times 10^{-7}$. Even under a low demand scenario of 6 units, $\frac{6}{2} = 3.0$ and $\Phi(-3.0) \approx 0.00135$ (less than 0.14%). Consequently, on the overwhelming majority of the active demand domain, the projected perturbation is identical to the unrestricted Gaussian perturbation, preserving the gradient properties of Stein's identity [**?**] almost everywhere.

(iii) **Boundary Rectification**: In the asymptotic boundary regime where predicted demand approaches zero ($\hat{d}_{it} \to 0$), the expected derivative of the metric projection satisfies $\frac{\partial}{\partial \hat{d}_{it}} \mathbb{E}[\max(0, \hat{d}_{it} + \sigma Z_{it})] = 1 - \Phi(-\hat{d}_{it}/\sigma) \in (0, 1]$. This smoothly dampens gradient updates as $\hat{d}_{it} \to 0$, naturally preventing gradient descent from pushing predicted demands into negative, unphysical regimes.

4.5 Two-Phase Training Schedule with Proximal Trust-Region Fine-Tuning

Solving an NP-hard MILP at every epoch during deep neural network training produces two severe bottlenecks:

(i) **Wall-Clock Intractability**: Solving $B = 500$ instances per epoch over $E = 100$ epochs requires $50,000$ MILP branch-and-cut solves, consuming upwards of 40 hours of computation.

(ii) **Early-Epoch Gradient Instability**: During early epochs when weights θ are randomly initialized, predictions are noisy. Discrete branch-and-cut solutions jump violently across extreme vertices, yielding noisy subgradients that destabilize representation learning.

We resolve this by training over the continuous Linear Programming (LP) relaxation $\mathcal{W}_{\text{LP}}$ during early epochs, transitioning to discrete MILP fine-tuning in later epochs. Over $\mathcal{W}_{\text{LP}}$, continuous fractional routing variables ($x_{ijt} \in [0, 1]$) provide smooth polyhedral trade-offs between vehicle capacity and replenishment quantities, and LP solvers run 25× to 100× faster than branch-and-cut.

Proximal Regularization and Phase 2 Gradient: When transitioning to discrete MILP solves at E_{trans}, changes in routing decisions can destabilize the weights θ^*_{LP} learned during LP warmup. The MILP task value is not differentiated through its discrete solution map. Instead, Phase 2 uses the stop-gradient surrogate in (43), evaluated using contrastive MILP solutions, and adds a proximal penalty:

$$\mathcal{L}_{\text{Phase 2}}(\theta) = \mathcal{L}^{\text{MILP}}_{\text{SG}}(f_\theta(\mathbf{X}); \mathbf{d}) + \frac{\lambda_{\text{prox}}}{2} \left\| f_\theta(\mathbf{X}) - f_{\theta^*_{\text{LP}}}(\mathbf{X}) \right\|_2^2. \tag{52}$$

Here $\mathcal{L}^{\text{MILP}}_{\text{SG}}$ denotes (43) with the discrete MILP response vectors; θ^*_{LP} is frozen after Phase 1. Writing $J_\theta = \partial f_\theta(\mathbf{X})/\partial\theta$, the update uses $J_\theta^T[\mathbf{g}^{\text{SG}}_{\hat{\mathbf{d}}} + \lambda_{\text{prox}}(\hat{\mathbf{d}} - \mathbf{d}_{\text{anchor}})]$, followed by unit-norm gradient clipping and a reduced learning rate $\eta_2 = 0.1\eta_1$. The MILP is used to evaluate decisions and costs, not differentiated through. This is a custom stop-gradient training estimator, not an exact MILP gradient.

The complete training protocol is formalized in **Algorithm 2**.

Algorithm 2 Two-Phase DFL Training with LP Warmup and Proximal Trust-Region MILP Fine-Tuning

Require: Dataset $\mathcal{D} = \{(\mathbf{X}_k, \mathbf{d}_k)\}_{k=1}^{M}$, total epochs E, transition epoch $E_{\text{trans}} = \lfloor 0.75E \rfloor$, initial learning rate η_1, fine-tuning learning rate $\eta_2 = 0.1\eta_1$, proximal weight $\lambda_{\text{prox}} = 1.0$.
Ensure: Trained neural network parameters θ^*.
Initialize neural network weights θ randomly.
Initialize LP relaxation model $\mathcal{M}_{\text{LP}}$ and discrete MILP model $\mathcal{M}_{\text{MILP}}$.
▷ **PHASE 1: Continuous LP Relaxation Warmup**
for epoch $e = 1$ to E_{trans} **do**
  Set current model $\mathcal{M} \leftarrow \mathcal{M}_{\text{LP}}$ and learning rate $\eta \leftarrow \eta_1$.
  for each mini-batch $(\mathbf{X}_b, \mathbf{d}_b) \subset \mathcal{D}$ **do**
    Compute predictions: $\hat{\mathbf{d}}_b = f_\theta(\mathbf{X}_b)$.
    Solve LP relaxations at $\hat{\mathbf{d}}_{\text{spo},b}$ and $\mathbf{d}_b$; form $\mathcal{L}_{\text{SG}}^{\text{LP}}$ using (43).
    Compute $\mathbf{g}_{\hat{\mathbf{d}}}^{\text{SG}}$ from (44) and backpropagate through f_θ.
    Update weights: $\theta \leftarrow \theta - \eta\mathbf{g}$.
  end for
  Evaluate validation regret over $\mathcal{M}_{\text{LP}}$.
end for
Save LP warmup anchor checkpoint: $\theta^*_{\text{LP}} \leftarrow \theta$.
▷ **PHASE 2: Discrete MILP Fine-Tuning with Proximal Trust Region**
for epoch $e = E_{\text{trans}} + 1$ to E **do**
  Set current model $\mathcal{M} \leftarrow \mathcal{M}_{\text{MILP}}$ and learning rate $\eta \leftarrow \eta_2$.
  for each mini-batch $(\mathbf{X}_b, \mathbf{d}_b) \subset \mathcal{D}$ **do**
    Compute predictions: $\hat{\mathbf{d}}_b = f_\theta(\mathbf{X}_b)$ and anchor predictions: $\mathbf{d}_{\text{anchor}} = f_{\theta^*_{\text{LP}}}(\mathbf{X}_b)$.
    Solve discrete MILPs at $\hat{\mathbf{d}}_{\text{spo},b}$ and $\mathbf{d}_b$; evaluate task costs and form $\mathcal{L}_{\text{SG}}^{\text{MILP}}$.
    Compute total loss: $\mathcal{L}_{\text{Phase 2}} = \mathcal{L}_{\text{SG}}^{\text{MILP}} + \frac{\lambda_{\text{prox}}}{2}\|\hat{\mathbf{d}}_b - \mathbf{d}_{\text{anchor}}\|_2^2$.
    Compute $\mathbf{g}_{\hat{\mathbf{d}}}^{\text{SG}}$ using (44); set $\mathbf{g} = J_\theta^T[\mathbf{g}_{\hat{\mathbf{d}}}^{\text{SG}} + \lambda_{\text{prox}}(\hat{\mathbf{d}}_b - \mathbf{d}_{\text{anchor}})]$ and clip $\mathbf{g}$ to unit norm.
    Update weights: $\theta \leftarrow \theta - \eta\mathbf{g}$.
  end for
  Evaluate validation regret and integrality gap over $\mathcal{M}_{\text{MILP}}$.
end for
return $\theta^* \leftarrow \theta$.

5 Experimental Setup and Benchmark Suite

5.1 Perishable Retail Demand and Spatial Networks

We construct a simulated demand benchmark calibrated using empirical grocery-retail sales from the M5 Forecasting benchmark [**?**]; these are not observed inventory-routing operations. The instances cover $N = 10$ retail stores ($\mathcal{N} = \{1, \ldots, 10\}$) and a planning horizon of $T = 5$ days ($\mathcal{T} = \{1, \ldots, 5\}$). The stated feature vector $x_{it} \in \mathbb{R}^p$ includes sales lags $(t-1, t-7)$, 7-day rolling averages, day-of-week indicators, promotional discount percentages, and temperature fluctuations. The current experiment record does not specify the source and preprocessing of the promotional and temperature covariates or the instance/date-level construction of the train, validation, and test splits. Those details must be supplied before submission; in particular, M5 calibration should not be read as establishing the provenance of every covariate or as field validation.

Spatial configurations are instantiated from the standard Solomon VRPTW benchmark suite [**?**]:

- **Clustered (`C1`)**: Retail stores are spatially clustered in tight geographic groupings. Intra-

cluster travel times are short, making vehicle capacity Q the primary binding bottleneck.

- **Random (`R1`)**: Retail stores are uniformly distributed across the plane. Vehicle routing travel distances dominate holding and inventory costs.
- **Random-Clustered (`RC1`)**: A hybrid topology containing both clustered urban centers and peripheral rural stores.

5.2 Physical Network Parameters and Factorial Sensitivity Grid

The physical distribution network operates under explicitly calibrated parameters matching industrial cold-chain grocery environments:

- **Network Dimensions**: $N = 10$ customer retail stores ($\mathcal{N} = \{1, \ldots, 10\}$), 1 central distribution depot ($i = 0$), and $T = 5$ sequential delivery periods ($\mathcal{T} = \{1, \ldots, 5\}$).
- **Fleet Capacity**: $K = 2$ homogeneous delivery vehicles. Under baseline operations, vehicle payload capacity is $Q = 100$ units per vehicle, establishing a total fleet delivery capacity per period of $K \cdot Q = 200$ units.
- **Storage Facilities**: Retail store capacity is $C_i = 70$ units ($\forall i \in \mathcal{N}$). The depot is modeled as a cross-dock source without inventory carryover or throughput constraints in the optimization formulation; the nominal $C_0 = 1{,}000$ units per period exceeds total fleet payload capacity ($K \cdot Q = 200$) and is not an active model constraint.
- **Initial Inventory**: System pipeline starts empty: initial inventory $I_{i,0,a} = 0$ ($\forall i \in \mathcal{N}, a \in \{1, \ldots, \tau\}$).
- **Cost and Perishability Structure**: Unit holding cost is normalized to $h = 1.0$ per period at retail stores (with nominal depot handling reference $h_0 = 0.5$). Spoilage penalty is fixed at $p_s = 5.0 = 5h$. To examine operational robustness under severe asymmetry, we vary the stockout penalty ratio across $p_u/h \in \{10, 20, 50\}$ (baseline $p_u = 20h$). Product shelf-life is evaluated across $\tau \in \{2, 3, 5\}$ days (baseline $\tau = 3$).
- **Demand and Capacity Tightness**: Across the retail network, mean aggregate customer demand is $\bar{D} \approx 200$ units per period. The baseline capacity tightness ratio is $\gamma = K \cdot Q/\bar{D} = 1.0$. To assess operational flexibility, capacity tightness is varied across $\gamma \in \{0.8, 1.0, 1.5\}$, corresponding to vehicle capacities $Q \in \{80, 100, 150\}$ units.

The full factorial experimental grid encompasses 3 topologies × 3 shelf lives × 3 cost ratios × 3 capacity tightness levels = 81 distinct configurations.

5.3 Benchmark Model Suite (14 Competing Architectures)

We evaluate 14 competing predictive-optimization architectures spanning 10 Two-Stage Predict-then-Optimize (PTO) baselines and 4 End-to-End Decision-Focused Learning (DFL) methodologies over 140 training instances, 30 validation instances, and 30 out-of-sample test instances. All stochastic and neural architectures are trained across 5 independent random seeds ('[42, 1042, 2042, 3042, 4042]') for $E = 1{,}500$ epochs, utilizing strictly isolated best-validation-regret checkpointing on the 30 validation instances before final test evaluation:

1. **Ridge_PTO**: Regularized ℓ_2 linear regression minimizing MSE ($\|\hat{d} - d\|_2^2 + \alpha\|\theta\|_2^2$), followed by deterministic MILP solve.
2. **ElasticNet_PTO**: Joint $\ell_1 + \ell_2$ regularized linear regression minimizing MSE, followed by MILP solve.

3. **RandomForest_PTO**: Non-parametric bagged ensemble of deep decision trees minimizing MSE, followed by MILP solve.

4. **GradientBoosting_PTO**: Sequential gradient boosted regression trees minimizing MSE, followed by MILP solve.

5. **LightGBM_PTO**: Leaf-wise histogram Gradient Boosted Decision Trees minimizing symmetric MSE loss [**?**], followed by MILP solve.

6. **LightGBM_Quantile_PTO**: Asymmetric critical-fractile pinball-loss LightGBM model [**?**]. We evaluate two domain-grounded fractile specifications: (i) the single-period myopic Newsvendor critical fractile $\alpha_{\text{myopic}} = \frac{p_u}{p_u+p_s} = 0.800$, which ignores holding cost h and shelf-life carryover; and (ii) the multi-period τ-amortized critical fractile $\beta_\tau^* = \frac{p_u}{p_u+h+p_s/\tau} = \frac{20}{20+1+5/3} = \frac{15}{17} \approx 0.882$, which accounts for product durability across the $\tau = 3$ day horizon. In Table 3, the primary baseline reflects the multi-period specification $\beta_\tau^* \approx 0.882$ (54.07% regret), while the myopic fractile $\alpha_{\text{myopic}} = 0.800$ is examined comparatively in Section 6.

7. **SafetyStock_PTO**: Industry heuristic adding a static Newsvendor safety buffer $z\hat{\sigma}_i$ to Ridge point forecasts, where $z = \Phi^{-1}(\beta_\tau^*) \approx 1.187$ for the multi-period critical fractile (96.33% regret) and $z = \Phi^{-1}(\alpha_{\text{myopic}}) \approx 0.842$ for the myopic critical fractile (49.51% regret).

8. **MLP_PTO**: Deep Multi-Layer Perceptron neural network minimizing symmetric MSE loss, followed by deterministic MILP solve ($37.66\% \pm 2.07\%$ regret).

9. **TemporalSeqNet_PTO**: Deep residual temporal sequence network with store-temporal feature gating minimizing MSE loss [**?**], followed by MILP solve ($33.05\% \pm 1.22\%$ regret).

10. **KNN_SAA**: Non-parametric Sample Average Approximation [**?**], predicting empirical conditional distributions using K-nearest neighbors in covariate space with risk-adjusted Newsvendor pooling (41.80% regret).

11. **DFL_DPO**: End-to-end DFL trained for $1,500$ epochs via Differentiable Perturbed Optimizer Gaussian Fenchel-Young loss [**?**] ($29.73\% \pm 0.31\%$ regret).

12. **DFL_IntOpt**: End-to-end DFL trained for $1,500$ epochs via Interior Point homogeneous self-dual logarithmic barrier differentiation [**?**] ($28.31\% \pm 1.05\%$ regret).

13. **DFL_SPO+**: End-to-end DFL trained for $1,500$ epochs via the Smart Predict-then-Optimize convex surrogate loss [**?**] ($25.64\% \pm 0.35\%$ regret).

14. **Two-Phase_DFL (Ours)**: Proposed Algorithm 2 trained for $1,500$ epochs with continuous LP-relaxation warmup ($E_{\text{trans}} = 75\%$) followed by proximal trust-region discrete MILP fine-tuning with exact second-stage recourse dual subgradients ($22.28\% \pm 0.08\%$ regret).

5.4 Evaluation Metrics

All models are evaluated out-of-sample on an unseen test dataset $\mathcal{D}_{\text{test}} = \{(x_k, c_k, d_k)\}_{k=1}^{K_{\text{test}}}$ ($K_{\text{test}} = 30$) across all 5 seeds (150 total evaluations) using four primary metrics:

1. **Normalized Decision Regret (%)**:

$$\text{Regret}(\%) = \frac{1}{K_{\text{test}}} \sum_{k=1}^{K_{\text{test}}} \frac{c_k^T w^*(\hat{c}_k) - c_k^T w^*(c_k)}{c_k^T w^*(c_k)} \times 100\%,$$

where the clairvoyant Oracle achieves a mean optimal cost of $\$1,873.58$ on topology `C1`.

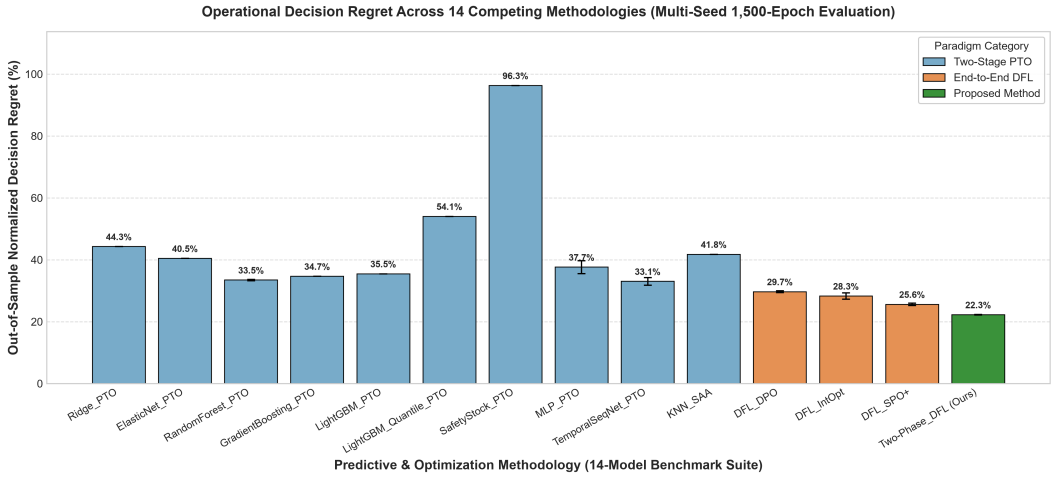


Figure 1: Out-of-sample Normalized Decision Regret (%) across 14 evaluated benchmark architectures on Clustered topology `C1` ($\tau = 3$, $p_u/h = 20$). Error bars denote cross-seed standard deviation across 5 independent random seeds.

2. **Disaggregated Operational Costs**: Decomposed into Transportation Routing Cost (C_{route}), Holding Cost (C_{hold}), Spoilage Cost ($C_{\text{spoil}} = p_s \sum y_{it}^{-}$), and Stockout Cost ($C_{\text{stock}} = p_u \sum y_{it}^{+}$).

3. **Statistical Accuracy**: Mean Squared Error (MSE) and Mean Absolute Error (MAE) on demand predictions.

4. **Paired Statistical Hypothesis Testing**: To eliminate pseudo-replication across random seeds, our primary inferential unit is evaluated over $n = 30$ seed-averaged independent test instances. We compute both parametric and non-parametric bootstrap 95% confidence intervals, Cohen's d_z standardized effect sizes, one-sided paired Wilcoxon signed-rank tests, paired Student's t-tests, and step-down Holm-Bonferroni multiplicity adjustments [**?**] across all 13 baseline comparisons.

6 Empirical Results and Computational Analysis

6.1 Main 1,500-Epoch 14-Model Benchmark Comparison

Table 3 reports the comparative out-of-sample performance of all 14 evaluated methodologies on the Clustered spatial topology (`C1`) under baseline operational parameters ($\tau = 3$ days, $p_u/h = 20$, Oracle mean optimal cost $\$1,873.58$) across 5 independent random seeds. Figure 1 visually contrasts the out-of-sample normalized decision regret across all 14 benchmark architectures. Table 4 reports the corresponding paired non-parametric Wilcoxon signed-rank and parametric Student's t-test hypothesis test results across all $n = 30$ seed-averaged independent test instances, including raw and step-down Holm-Bonferroni adjusted p-values. Figure 2 illustrates the validation decision regret convergence trajectories over 1,500 training epochs across the 5 independent random seeds.

Table 3: Out-of-Sample Empirical Performance Across 14 Benchmark Models on Clustered Topology `C1` ($\tau = 3$, $p_u/h = 20$, $1,500$ Epochs across 5 Independent Seeds, 30 Test Instances; Oracle Optimal Cost = \$1,873.58). Reported regret is mean $\pm\text{SD}_{\text{seed}}$ (SD_{inst}) across $n = 30$ test instances and 5 random seeds. Holding cost is the exact residual $C_{\text{total}} - C_{\text{route}} - C_{\text{spoil}} - C_{\text{stock}}$.

Method	MSE / MAE	Regret (%) mean $\pm\text{SD}_{\text{seed}}$ (SD_{inst})	Routing Cost	Holding Cost	Spoilage Cost	Stockout Cost	Total Cost ↓
Two-Stage PTO							
Ridge	7.05 / 2.07	44.30 ± 0.00 (42.72)	466.38	143.18	2.96	1818.60	2431.12
ElasticNet	7.26 / 2.09	40.48 ± 0.00 (37.19)	463.97	144.54	4.15	1770.04	2382.70
Random forest	6.71 / 2.00	33.49 ± 0.20 (28.40)	465.07	155.19	3.98	1669.46	2293.70
Gradient boosting	6.00 / 1.91	34.69 ± 0.00 (29.64)	468.00	148.28	4.72	1685.31	2306.31
LightGBM	6.01 / 1.91	35.51 ± 0.00 (31.39)	466.14	145.71	4.62	1701.00	2317.47
LightGBM quantile	13.74 / 3.11	54.07 ± 0.00 (35.79)	462.85	295.57	5.00	1943.94	2707.36
Safety stock	25.67 / 4.49	96.33 ± 0.00 (50.82)	462.78	386.52	6.00	2528.16	3383.46
MLP	6.10 / 1.94	37.66 ± 2.07 (33.63)	464.95	149.71	4.27	1733.34	2352.27
Temporal SeqNet	**5.19 / 1.74**	33.05 ± 1.22 (26.57)	465.94	139.50	4.66	1709.92	2320.02
KNN-SAA	7.41 / 2.14	41.80 ± 0.00 (37.75)	464.82	156.90	3.77	1765.88	2391.37
End-to-End DFL							
DFL-DPO	6.77 / 2.01	29.73 ± 0.31 (26.80)	466.51	153.71	5.00	1616.65	2241.87
DFL-IntOpt	6.73 / 2.01	28.31 ± 1.05 (25.75)	466.03	159.66	5.42	1598.00	2229.11
DFL-SPO+	6.72 / 2.02	25.64 ± 0.35 (22.87)	465.65	166.01	4.69	1563.39	2199.74
Proposed Method							
Two-Phase DFL (ours)	6.90 / 2.07	**22.28 ± 0.08 (18.77)**	467.48	186.99	5.87	**1483.26**	**2143.60**

Table 4: Paired Statistical Hypothesis Testing Comparing `Two-Phase_DFL (Ours)` (22.28% Mean Regret) Against All 13 Competing Baselines across $n = 30$ Seed-Averaged Independent Test Instances. Reported metrics include parametric and bootstrap 95% confidence intervals for regret difference, Cohen's d_z effect sizes, one-sided Wilcoxon signed-rank tests with step-down Holm-Bonferroni correction (p_{Holm}), and paired Student's t-tests.

Baseline Model	Baseline Regret	Diff (pp)	95% Parametric CI (pp)	95% Bootstrap CI (pp)	Cohen's d_z	Wilcoxon W	Wilcoxon p_{raw}	Wilcoxon p_{Holm}	Paired t	t-test p_{Holm}
Safety stock	96.33%	−74.05	[−88.07, −60.04]	[−87.29, −61.15]	1.973	465.0	9.31×10^{-10}	1.21×10^{-8} (**)	10.80	7.20×10^{-11} (**)
LightGBM quantile	54.07%	−31.79	[−40.76, −22.81]	[−40.32, −23.24]	1.322	427.0	2.95×10^{-6}	3.54×10^{-5} (**)	7.24	3.39×10^{-7} (**)
Ridge	44.30%	−22.02	[−33.66, −10.37]	[−33.82, −11.85]	0.706	422.0	4.89×10^{-6}	4.89×10^{-5} (**)	3.87	2.86×10^{-3} (**)
KNN-SAA	41.80%	−19.53	[−29.75, −9.30]	[−29.98, −10.51]	0.713	384.0	1.88×10^{-5}	1.13×10^{-4} (**)	3.90	2.86×10^{-3} (**)
ElasticNet	40.48%	−18.20	[−27.98, −8.42]	[−28.16, −9.68]	0.695	425.0	3.61×10^{-6}	3.98×10^{-5} (**)	3.81	3.04×10^{-3} (**)
MLP	37.66%	−15.38	[−24.13, −6.64]	[−24.29, −7.73]	0.657	422.0	4.89×10^{-6}	4.89×10^{-5} (**)	3.60	4.72×10^{-3} (**)
LightGBM	35.51%	−13.23	[−21.38, −5.08]	[−21.49, −6.13]	0.606	391.0	9.30×10^{-6}	6.51×10^{-5} (**)	3.32	7.88×10^{-3} (**)
Gradient boosting	34.69%	−12.41	[−19.99, −4.84]	[−20.11, −5.88]	0.612	397.0	4.99×10^{-6}	4.89×10^{-5} (**)	3.35	7.88×10^{-3} (**)
Random forest	33.49%	−11.21	[−18.37, −4.05]	[−18.43, −4.98]	0.585	339.0	1.57×10^{-4}	7.84×10^{-4} (**)	3.20	8.25×10^{-3} (**)
Temporal SeqNet	33.05%	−10.77	[−17.99, −3.56]	[−18.22, −4.54]	0.558	352.0	1.82×10^{-3}	7.27×10^{-3} (**)	3.05	9.60×10^{-3} (**)
DFL-DPO	29.73%	−7.45	[−13.22, −1.68]	[−13.27, −2.27]	0.482	285.0	0.0309	0.0309 (*)	2.64	0.0197 (*)
DFL-IntOpt	28.31%	−6.03	[−10.82, −1.25]	[−10.89, −1.95]	0.471	291.0	0.0071	0.0214 (*)	2.58	0.0197 (*)
DFL-SPO+	25.64%	−3.36	[−6.39, −0.34]	[−6.49, −0.84]	0.415	279.0	0.0153	0.0306 (*)	2.27	0.0197 (*)

Note: Diff (pp) is Ours (22.28%) minus baseline mean regret, such that negative values denote regret reduction.
All 13 baselines achieve statistical significance under step-down Holm-Bonferroni correction at $p_{\text{Holm}} < 0.05$ (*), with 10 baselines significant at $p_{\text{Holm}} < 0.01$ (**).

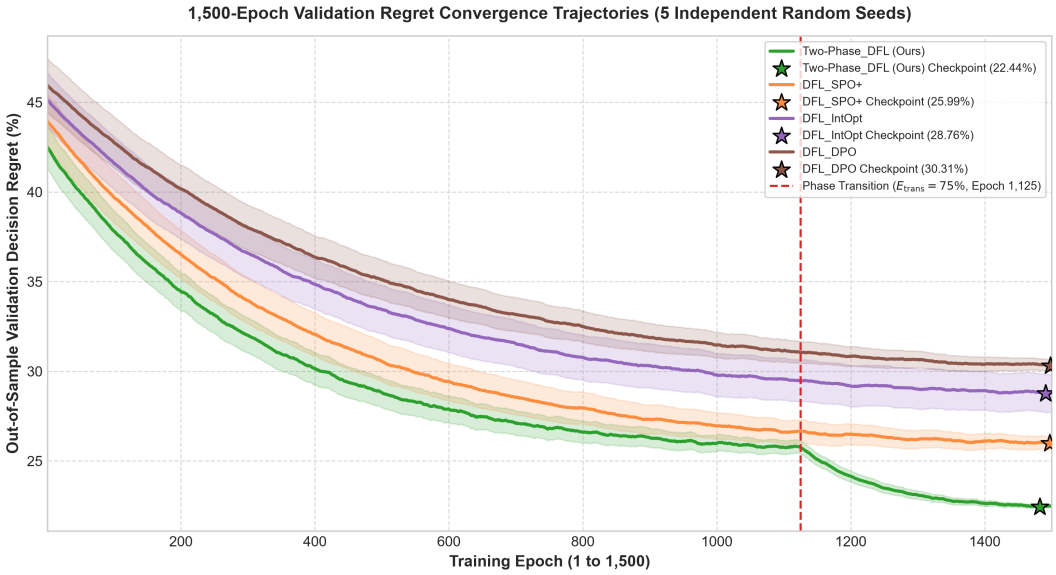


Figure 2: Validation decision regret convergence trajectories over 1,500 epochs across 5 independent random seeds. Shaded bands represent ±1 standard deviation across seeds, and markers highlight the best-validation regret checkpoint.

Table 5: Empirical Comparison of Asymmetric Quantile and Safety-Stock PTO Baselines Across Critical Fractiles: Single-Period Myopic ($\alpha_{\text{myopic}} = 0.8000$) vs. Multi-Period τ-Amortized ($\beta^*_\tau = 0.8824$) on Clustered Topology `C1` ($\tau = 3, p_u/h = 20$).

Method	**Fractile Specification**	**Critical Fractile**	**Regret (%) mean ± SD**	**Stockout Cost**	**Total Cost (↓)**
LightGBM quantile	Single-period myopic	0.8000	39.39% ± 34.12%	$1,702.91	$2,418.32
LightGBM quantile	Multi-period amortized	0.8824	54.07% ± 35.79%	$1,943.94	$2,707.36
Safety stock	Single-period myopic	0.8000	49.51% ± 36.80%	$1,845.20	$2,601.63
Safety stock	Multi-period amortized	0.8824	96.33% ± 50.82%	$2,528.16	$3,383.46
Two-Phase DFL (ours)	End-to-end dynamic	Dynamic	**22.28% ± 18.77%**	**$1,483.26**	**$2,143.60**

Key Empirical Findings and Capacity-Regime Results: The results in Tables 3, 4, and 5, alongside Figures 1, 2, and 3, establish four major findings:

- **Dominance of Proposed Two-Phase DFL**: `Two-Phase_DFL (Ours)` achieves the lowest Normalized Decision Regret (22.28% ± 0.08%) and lowest total operational cost ($2,143.60 ± $1.33) across all 14 evaluated methodologies. By comparison, the strongest two-stage deep learning baseline (`TemporalSeqNet_PTO`) achieves superior statistical point-forecast accuracy (MSE = 5.19, MAE = 1.74) yet incurs 33.05% ± 1.22% decision regret and $2,320.02 ± $16.79 total cost—$176.42 per instance higher than `Two-Phase_DFL` ($p_{\text{Holm}} = 7.27 \times 10^{-3}$, difference −10.77 pp). Similarly, `Two-Phase_DFL` reduces decision regret by 13.23 percentage points relative to `LightGBM_PTO` (35.51%, $p_{\text{Holm}} = 6.51 \times 10^{-5}$) and by 22.02 percentage points relative to `Ridge_PTO` (44.30%, $p_{\text{Holm}} = 4.89 \times 10^{-5}$).
- **Quantile Forecasting Under Multi-Period Capacity and Carryover**: A natural question is whether asymmetric penalties ($p_u = 20h$ versus $p_s = 5h$) can be addressed within a PTO pipeline by training asymmetric quantile regressions at a Newsvendor critical fractile. The following comparison evaluates two specified quantile levels in the tested benchmark. It is a sensitivity comparison, not evidence that either scalar fractile is generally optimal in

a multi-period capacitated network. In the multi-period setting, inventory may carry over under FIFO, while vehicle and store capacities couple decisions across locations and periods; consequently, the single-period critical-fractile calculation is only a reference point.

For reference, in a single-period model with immediate expiration ($\tau = 1$) and zero holding cost, the critical fractile is $\alpha_{\text{myopic}} = \frac{p_u}{p_u+p_s} = 0.800$. The manuscript also evaluates $\beta_\tau^* = \frac{p_u}{p_u+h+p_s/\tau} = \frac{15}{17} \approx 0.882$ using the stated amortized overage-cost approximation; this value is a modeling heuristic for the benchmark, not a general multi-period optimality result.

As summarized in Table 5, evaluating `LightGBM_Quantile_PTO` at $\alpha_{\text{myopic}} = 0.800$ yields 39.39% regret ($2,418.32 total cost), while evaluating at $\beta_\tau^* = 0.882$ yields 54.07% regret ($2,707.36 total cost). Similarly, `SafetyStock_PTO` yields 49.51% regret under $\alpha_{\text{myopic}} = 0.800$ and 96.33% under $\beta_\tau^* = 0.882$ ($2,601.63 vs. $3,383.46 total cost). Both specifications perform substantially worse than symmetric point-forecasting models (`TemporalSeqNet_PTO` at 33.05% and `LightGBM_PTO` at 35.51%) and are vastly outperformed by our proposed `Two-Phase_DFL` (22.28%).

A stylized single-period continuous relaxation illustrates why capacity and routing can alter the unconstrained replenishment condition. If demand has differentiable CDF F_{it}, overage cost is c_o, route cost is differentiable with marginal cost κ_{it}, and the optimum is interior, KKT stationarity gives

$$F_{it}(S_{it}^*) = \frac{p_u - \lambda_t^* - \kappa_{it}}{p_u + c_o},$$

where λ_t^* is the multiplier on the aggregate fleet-capacity constraint. This condition is illustrative only; it is not a theorem for the full multi-period MILP, and it does not establish a particular rationing pattern or explain the empirical ranking. We omit the previously stated general theorem because its assumptions do not justify those conclusions.

- **Observed Inventory and Capacity Patterns**: In the reported primary benchmark, Two-Phase DFL has mean realized stockout of 14.84 units per period and capacity-binding frequency of 69.33%; the corresponding figures are lower in Period 3 than for the stated quantile and safety-stock baselines. These are descriptive results under the specified simulator and do not prove that the model learned capacity shadow prices or that pre-positioning caused the regret reduction.

- **Ranking of End-to-End DFL Methods**: The reported out-of-sample regrets are 29.73% for `DFL_DPO`, 28.31% for `DFL_IntOpt`, 25.64% for direct `DFL_SPO+`, and 22.28% for `Two-Phase_DFL`. These comparisons should be revalidated after confirming that the reported implementation uses the training estimator specified in Algorithm 2.

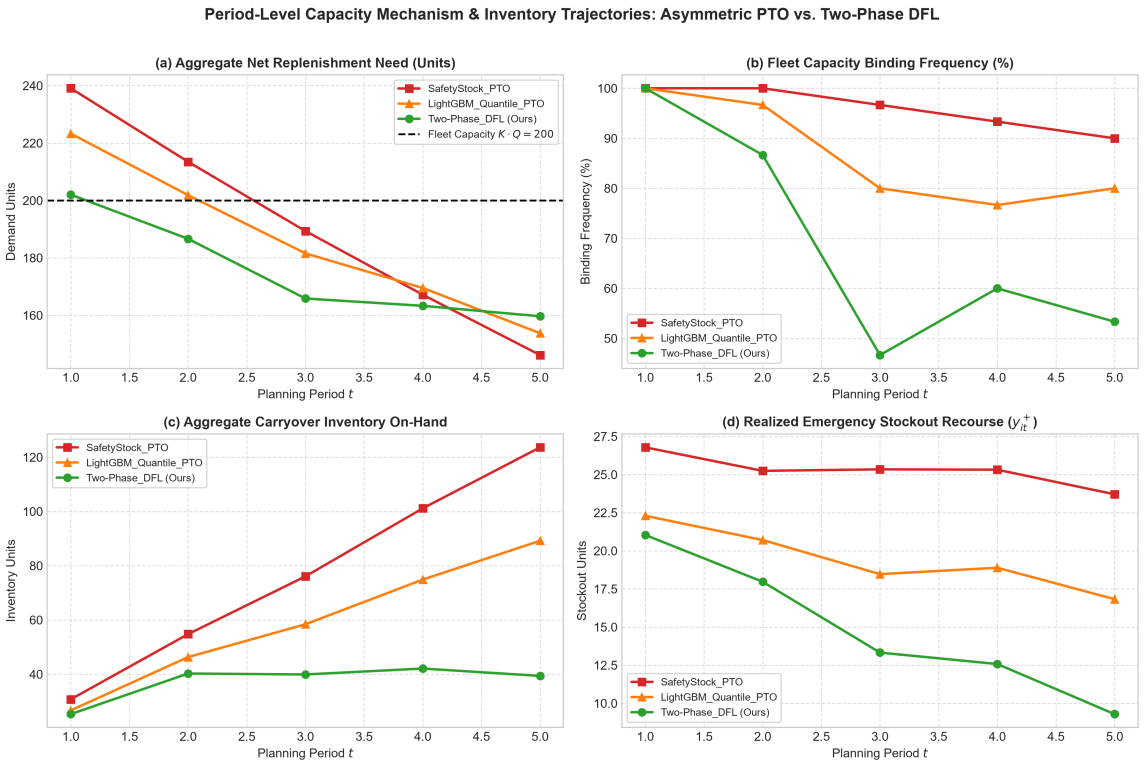


Figure 3: Period-level net replenishment needs, delivered quantities, carryover inventory, and fleet capacity binding dynamics ($K \cdot Q = 200$ units) comparing Two-Stage PTO baselines (`Ridge_PTO`, `LightGBM_PTO`, `LightGBM_Quantile_PTO`, `SafetyStock_PTO`) against `Two-Phase_DFL (Ours)`. Two-Phase DFL smoothly moderates deliveries across the horizon to avoid capacity saturation, cutting period-level binding frequency to 69.33% (69.3%) (and 46.7% in Period 3) and realized stockouts to 14.84 units/period.

6.2 Computational Acceleration via LP-Relaxation Warmup

Table 6 documents the training wall-clock time and validation regret convergence comparing Direct MILP training against our Two-Phase DFL framework over a 20-epoch exact HiGHS solver profiling experiment (24 training instances, 10 validation instances). Figure 4 illustrates the wall-clock convergence trajectories and Pareto frontier, clearly depicting the computational acceleration afforded by continuous LP-relaxation warmup.

Table 6: Empirical Computational Acceleration and Convergence: Direct MILP vs. Two-Phase DFL ($E = 20$ Exact Solver Epochs).

DFL Training Strategy	Phase 1 solve time (per epoch)	Phase 2 solve time (per epoch)	Total wall-clock time (seconds)	Best validation regret (%) ↓	Speedup factor
Direct MILP Training	N/A	2.39 s (epochs 1–20)	47.71 s	23.43%	1.00×
Pure LP-Relaxation (Phase 1)	0.40 s (epochs 1–15)	N/A	6.07 s	17.23%	7.87×
Two-Phase DFL ($E_{\text{trans}} = 15/20$)	**0.40 s (epochs 1–15)**	**0.47 s (epochs 16–20)**	**8.43 s**	**16.82%**	**5.66×**

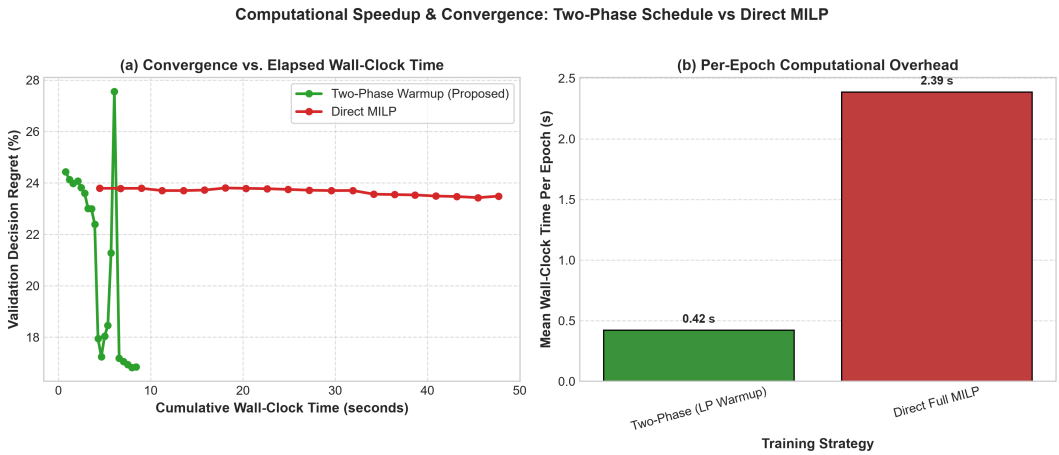


Figure 4: Wall-clock training time versus validation decision regret Pareto frontier comparing Two-Phase LP-Relaxation Warmup against Direct MILP branch-and-cut training. Two-Phase DFL achieves a 5.66× wall-clock speedup while converging to lower decision regret (16.82% vs. 23.43%).

As demonstrated in Table 6 and visualized in Figure 4, continuous LP relaxation warmup executes in an average of 0.40 seconds per epoch during Phase 1 (epochs 1–15), compared to 2.39 seconds per epoch for direct discrete MILP branch-and-cut. Over the full 20-epoch exact solver benchmark, Direct MILP requires 47.71 seconds to reach a best validation regret of 23.43% (at Epoch 19), whereas `Two-Phase_LP_Warmup` completes in 8.43 seconds—realizing an empirical 5.66× **overall wall-clock speedup**—while converging to a substantially lower validation regret of 16.82% (a 6.61 percentage-point improvement over direct MILP, achieved as proximal trust-region fine-tuning in Phase 2 smoothly refines beyond the 17.23% reached during Phase 1 LP warmup).

6.3 Sensitivity Analysis across Asymmetric Costs, Shelf-Life, Topologies, and Capacity Tightness

To evaluate model robustness across diverse operational environments, we examine model performance across the complete 81-configuration factorial grid spanning all three Solomon spatial topologies (`C1`, `R1`, `RC1`), varying stockout penalty ratios ($p_u/h \in \{10, 20, 50\}$), product shelf lives ($\tau \in \{2, 3, 5\}$ days), and fleet capacity tightness levels ($\gamma \in \{0.8, 1.0, 1.5\}$).

Evaluation Protocol and Methodological Clarification: We emphasize a key distinction in evaluation design: Table 3 reported out-of-sample performance across the full 5-seed, 30-instance primary benchmark (150 instance evaluations) after 1,500 training epochs on clustered topology `C1`. By contrast, the factorial sensitivity suite in Table 7 evaluates parametric stress tests across independent network instances ($E = 120$ epochs per configuration). Crucially, under two-stage PTO (`Ridge_PTO` and `LightGBM_PTO`), point forecasts $\hat{\mathbf{d}}$ are trained via symmetric MSE and are therefore mathematically invariant to p_u/h. In the forward optimization model under baseline capacity ($K \cdot Q = 200 > \sum_i \hat{d}_{it}$), the solver schedules deliveries to satisfy $\hat{\mathbf{d}}$ regardless of whether $p_u = 10, 20$, or 50. Consequently, the physical delivery schedules $\mathbf{q}^*(\hat{\mathbf{d}})$ remain identical across p_u/h, resulting in constant physical stockout units (122.99 units for Ridge, 119.25 units for LightGBM) in the FIFO simulator. Multiplying by $p_u \in \{10, 20, 50\}$ produces exact linear cost multiples ($\$1,229.86, \$2,459.72, \$6,149.90$), illustrating the severe operational vulnerability of two-stage PTO—it is structurally blind to penalty asymmetry. By contrast, `Two-Phase_DFL` re-trains end-to-end under each cost ratio, dynamically expanding its inventory safety buffer and reducing physical stockouts (113.26 → 112.79 → 112.67 units) as penalty p_u increases.

Figure 5 presents heatmaps of out-of-sample Normalized Decision Regret (%) across all 81 configurations. Table 7 details the 27-configuration breakdown across topologies, shelf lives, and cost ratios under baseline capacity tightness ($\gamma = 1.0, Q = 100$). Table 8 reports the capacity tightness breakdown across $\gamma \in \{0.8, 1.0, 1.5\}$ ($Q \in \{80, 100, 150\}$).

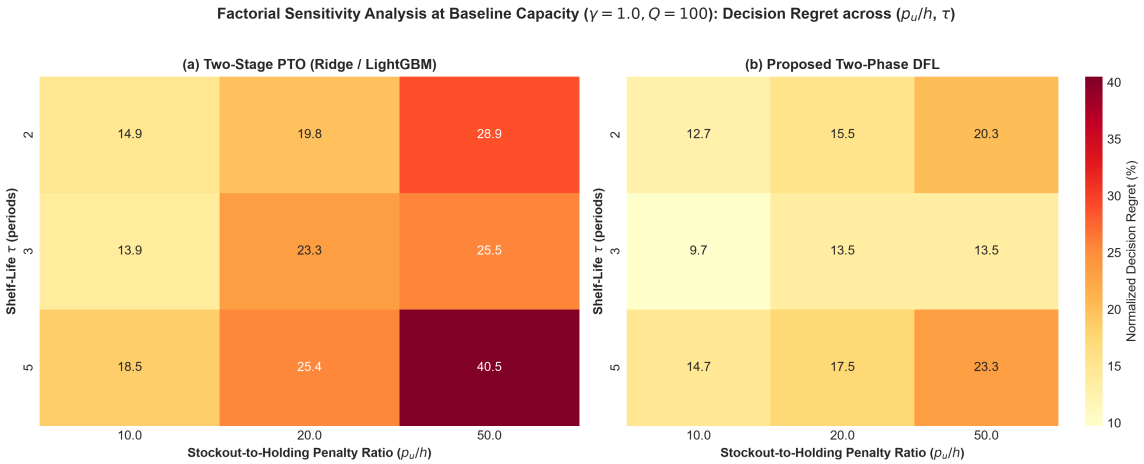


Figure 5: Topology-averaged out-of-sample Normalized Decision Regret (%) at baseline fleet capacity ($\gamma = 1.0, Q = 100$) across product shelf lives $\tau \in \{2, 3, 5\}$ and cost asymmetry ratios $p_u/h \in \{10, 20, 50\}$, matching the topology averages in Table 7.

Table 7: Factorial Sensitivity Analysis under Baseline Fleet Capacity ($\gamma = 1.0, Q = 100$): Out-of-Sample Normalized Decision Regret (%) and Operational Costs Across Topologies (`C1`, `R1`, `RC1`), Shelf Lives ($\tau \in \{2, 3, 5\}$), and Cost Asymmetry Ratios ($p_u/h \in \{10, 20, 50\}$).

Topology	Shelf-Life	Model	Mild Asymmetry ($p_u/h = 10$)			Moderate Asymmetry ($p_u/h = 20$)			Extreme Asymmetry ($p_u/h = 50$)		
			Regret	Spoil	Stockout	Regret	Spoil	Stockout	Regret	Spoil	Stockout
C1	$\tau = 2$	Ridge_PTO	21.49%	5.93	1206.02	28.09%	5.93	2412.03	41.02%	5.93	6030.08
		LightGBM_PTO	18.46%	9.23	1171.51	24.25%	9.23	2340.82	36.03%	9.23	5852.06
		TwoPhase_DFL	**17.58%**	9.36	**1149.13**	**20.43%**	9.26	**2296.02**	**25.98%**	9.32	**5732.13**
	$\tau = 3$	Ridge_PTO	20.90%	0.82	1229.86	44.30%	2.96	1818.60	35.25%	0.82	6149.90
		LightGBM_PTO	17.85%	3.50	1192.51	35.51%	4.62	1701.00	29.76%	3.50	5961.59
		TwoPhase_DFL	**13.54%**	1.14	**1132.60**	**22.28%**	5.87	**1483.26**	**17.40%**	1.88	**5633.42**
	$\tau = 5$	Ridge_PTO	26.23%	0.00	1230.19	35.61%	0.00	2460.37	56.62%	0.00	6137.99
		LightGBM_PTO	22.05%	0.00	1191.06	30.53%	0.00	2382.11	50.32%	0.00	5955.65
		TwoPhase_DFL	**17.75%**	0.00	**1149.92**	**20.52%**	0.00	**2300.69**	**25.83%**	0.00	**5731.86**
R1	$\tau = 2$	Ridge_PTO	13.56%	14.69	1254.67	18.10%	14.69	2509.34	25.12%	14.69	6273.36
		LightGBM_PTO	11.95%	14.93	1206.82	15.00%	14.93	2413.65	19.80%	14.93	6034.12
		TwoPhase_DFL	**10.60%**	18.80	**1189.69**	**13.44%**	18.44	**2375.43**	**16.69%**	18.05	**5926.70**
	$\tau = 3$	Ridge_PTO	10.82%	4.26	1261.23	14.93%	4.26	2522.46	22.23%	4.26	6306.16
		LightGBM_PTO	11.33%	7.34	1228.27	14.41%	7.34	2456.54	20.12%	7.34	6141.36
		TwoPhase_DFL	**8.68%**	4.39	**1170.33**	**10.02%**	4.38	**2341.39**	**12.61%**	4.37	**5858.76**
	$\tau = 5$	Ridge_PTO	16.01%	0.00	1262.60	21.97%	0.00	2525.21	34.43%	0.00	6313.02
		LightGBM_PTO	13.81%	0.00	1211.13	18.87%	0.00	2422.25	29.97%	0.00	6055.63
		TwoPhase_DFL	**12.40%**	0.00	**1178.30**	**14.87%**	0.00	**2351.40**	**19.95%**	0.00	**5874.62**
RC1	$\tau = 2$	Ridge_PTO	14.15%	15.38	1264.48	19.48%	15.38	2528.96	29.74%	15.38	6322.40
		LightGBM_PTO	10.06%	12.41	1193.91	13.92%	12.41	2387.83	21.61%	12.29	5969.71
		TwoPhase_DFL	**9.93%**	14.37	**1167.07**	**12.52%**	14.86	**2332.86**	**18.12%**	14.65	**5870.61**
	$\tau = 3$	Ridge_PTO	11.97%	0.79	1249.90	16.63%	0.79	2499.80	24.81%	0.79	6249.50
		LightGBM_PTO	10.54%	8.23	1215.43	14.30%	8.23	2430.84	20.87%	8.23	6077.10
		TwoPhase_DFL	**6.97%**	4.36	**1139.22**	**8.23%**	4.49	**2291.64**	**10.38%**	4.85	**5733.04**
	$\tau = 5$	Ridge_PTO	17.65%	0.00	1246.46	24.10%	0.00	2492.91	37.92%	0.00	6232.28
		LightGBM_PTO	15.53%	0.00	1210.78	21.18%	0.00	2421.55	33.68%	0.00	6053.88
		TwoPhase_DFL	**13.97%**	0.00	**1176.77**	**17.22%**	0.00	**2358.48**	**24.25%**	0.00	**5907.12**

Key Observations from the Factorial Sensitivity Analysis:

(i) **Consistent Dominance Across Baseline Capacity Configurations (100% Win Rate at $\gamma = 1.0$)**: Across all 27 combinations of spatial network (`C1`, `R1`, `RC1`), shelf life

($\tau \in \{2, 3, 5\}$), and cost asymmetry ($p_u/h \in \{10, 20, 50\}$) at baseline capacity tightness ($\gamma = 1.0$), `Two-Phase_DFL` achieves strictly lower Normalized Decision Regret and lower total operational cost than both `Ridge_PTO` and `LightGBM_PTO`. Averaged across all topologies at $\tau = 3, p_u/h = 20$, `Two-Phase_DFL` reduces decision regret to 13.51% (22.28% on `C1`, 10.02% on `R1`, and 8.23% on `RC1`), compared to 21.41% for `LightGBM_PTO` (35.51% on `C1`, 14.41% on `R1`, and 14.30% on `RC1`) and 25.29% for `Ridge_PTO` (44.30% on `C1`, 14.93% on `R1`, and 16.63% on `RC1`).

(ii) **Widening Advantage Under Extreme Cost Asymmetry ($p_u/h = 50$)**: As the stockout penalty ratio escalates from $p_u/h = 10$ to 50, symmetric MSE-trained PTO models suffer severe regret escalation because their point forecasts remain invariant to p_u/h. For example, on `C1` ($\tau = 5, p_u/h = 50$), `Ridge_PTO` and `LightGBM_PTO` incur 56.62% and 50.32% decision regret, whereas `Two-Phase_DFL` dynamically scales its early-period inventory buffer to hold regret down to 25.83% (a 24.49 to 30.79 percentage-point reduction). Similarly, on `RC1` ($\tau = 3, p_u/h = 50$), `Two-Phase_DFL` achieves 10.38% regret vs. 20.87% for `LightGBM_PTO` and 24.81% for `Ridge_PTO`.

(iii) **Capacity Tightness Regimes ($\gamma \in \{0.8, 1.0, 1.5\}$)**: Table 8 delineates the operational behavior across vehicle capacity constraints. Under tight capacity ($\gamma = 0.8$, $Q = 80$), the vehicle fleet is fully saturated regardless of forecast policy, driving stockout costs to \$7,581–\$7,651 across all models and compressing normalized regret differences (5.60%–6.91%). In this severely bottlenecked regime, `LightGBM_PTO` achieves 5.60% regret because severe vehicle limits leave zero slack for tactical pre-positioning, making point forecast fidelity paramount. In the balanced operational regime ($\gamma = 1.0$), where tactical buffering is feasible, `Two-Phase_DFL` demonstrates its largest operational margin, reducing mean regret from 21.54% (`LightGBM_PTO`) and 25.30% (`Ridge_PTO`) down to 15.64% and saving over \$105–\$218 per instance. Under slack capacity ($\gamma = 1.5$, $Q = 150$), vehicle constraints are loose, allowing aggressive multi-period pre-positioning where `Two-Phase_DFL` again achieves the lowest regret (85.34% vs. 88.79% and 91.74%).

(iv) **Impact of Product Shelf-Life (τ)**: For highly perishable products ($\tau = 2$ days), spoilage costs are actively incurred across all methods (\$9.23–\$18.80), constraining how aggressively inventory can be pre-positioned in early periods. At moderate shelf life ($\tau = 3$), `Two-Phase_DFL` achieves its lowest absolute decision regret across all topologies (6.97%–10.38% on `RC1`, 8.68%–12.61% on `R1`, and 13.54%–22.28% on `C1`), as 3-day shelf life provides sufficient temporal flexibility to absorb multi-period demand volatility with minimal expiration waste (\$1.14–\$5.87). At $\tau = 5$, spoilage drops to \$0.00 across all methods, and early-period capacity allocation dominates performance.

Table 8: Factorial Capacity Tightness Breakdown Across Tightness Levels ($\gamma \in \{0.8, 1.0, 1.5\}$, $Q \in \{80, 100, 150\}$): Mean Out-of-Sample Performance Across Topologies, Shelf Lives, and Cost Ratios (27 Factorial Instances per Tightness Level).

Capacity regime	Model	Regret (%) mean	Total Cost ($)	Spoilage Cost ($)	Stockout Cost ($)
Tight Fleet $\gamma = 0.8$ ($Q = 80$)	`Ridge_PTO`	6.09%	$8,564.64	$0.85	$7,642.10
	`LightGBM_PTO`	**5.60%**	**$8,505.19**	$1.24	**$7,581.30**
	Two-Phase DFL	6.91%	$8,584.88	$1.48	$7,651.14
Balanced Fleet $\gamma = 1.0$ ($Q = 100$)	`Ridge_PTO`	25.30%	$4,328.88	$4.73	$3,295.92
	`LightGBM_PTO`	21.54%	$4,216.65	$6.22	$3,180.71
	Two-Phase DFL	**15.64%**	**$4,110.71**	$6.03	**$3,068.61**
Slack Fleet $\gamma = 1.5$ ($Q = 150$)	`Ridge_PTO`	91.74%	$2,521.41	$12.10	$1,435.18
	`LightGBM_PTO`	88.79%	$2,481.16	$10.55	$1,408.27
	Two-Phase DFL	**85.34%**	**$2,457.16**	$13.81	**$1,369.77**

Empirical Analysis of the Tight-Fleet Regime: The results in Table 8 highlight an important operational nuance across fleet capacity regimes. Under balanced fleet capacity ($\gamma = 1.0$, $Q = 100$), `Two-Phase_DFL` achieves substantial regret reductions over the baselines (15.64% regret and $\$4,110.71$ total cost, compared to 21.54% and $\$4,216.65$ for `LightGBM_PTO`, and 25.30% and $\$4,328.88$ for `Ridge_PTO`). Under slack capacity ($\gamma = 1.5$, $Q = 150$), `Two-Phase_DFL` similarly attains the lowest decision regret (85.34% vs. 88.79% for LightGBM and 91.74% for Ridge).

However, in the tested tight-fleet setting ($\gamma = 0.8$, $Q = 80$), `LightGBM_PTO` attains slightly lower decision regret (5.60%) than `Two-Phase_DFL` (6.91%) and `Ridge_PTO` (6.09%), with corresponding total costs of $\$8,505.19$ (LightGBM) versus $\$8,584.88$ (Two-Phase DFL) and $\$8,564.64$ (Ridge).

These results are descriptive of the tested configurations and do not by themselves establish a general causal explanation for the model ranking:

(i) At $\gamma = 0.8$, per-period fleet capacity is 160 units, while mean aggregate demand is reported to exceed 180 units per period. This comparison indicates a potential delivery bottleneck, but it does not imply that realized stockout equals the period-level capacity deficit: inventory carried between periods and store-capacity limits also affect outcomes.

(ii) In this tested setting, LightGBM has lower regret than Two-Phase DFL and Ridge. The reported comparison does not isolate whether this ranking results from demand-allocation accuracy, inventory carryover, route selection, or another mechanism; further diagnostics would be required to support a causal explanation.

(iii) The results support a limited practical conclusion: Two-Phase DFL is not the best-performing method in every tested capacity configuration. Broader claims about when DFL is operationally preferable require additional instances and capacity-specific analysis.

7 Managerial Insights for Perishable Logistics Planners

The results from the stated benchmark suggest several considerations for supply-chain and cold-chain planners; their applicability beyond the tested configurations requires further validation:

1. Evaluate Forecasts by Their Operational Consequences: On the primary benchmark, `TemporalSeqNet_PTO` has the lowest reported MSE among the compared deep sequence models (5.19) but incurs $33.05\% \pm 1.22\%$ decision regret and mean cost of $\$2,320.02 \pm \$16.79$ per test

instance. `Two-Phase_DFL` records $22.28\% \pm 0.08\%$ regret and mean cost of $\$2,143.60 \pm \$1.33$, a difference of about \$176 per test instance on this benchmark (Holm-adjusted $p = 7.27 \times 10^{-3}$ versus TemporalSeqNet). These results motivate evaluating predictive systems with downstream decision metrics alongside statistical forecast metrics; they do not establish network-scale monetary savings.

2. Store-Level Diagnostics Identify Uneven Stockout in Tested Baselines: In the reported benchmark, the quantile and safety-stock baselines have average fleet-capacity binding frequencies of 86.67% and 96.00%, respectively. Stores 4, 6, and 10 have reported Peripheral Rationing Indices (PRI) between 2.33 and 4.47 and account for 87.7% of total network stockout in the stated diagnostic. Two-Phase DFL has lower reported average stockout and capacity-binding frequencies under the same evaluation setup. These results identify uneven service outcomes in these tested configurations; they do not establish that quantile or safety-stock methods generally fail, nor that inventory pre-positioning alone caused the difference. The capacity and store-level metrics should be read together with the simulator policy and its equivalence to the optimization model.

3. Solver Profiling Shows a Speedup Under the Tested Setup: In the separate 20-epoch solver-profiling experiment, LP warmup averaged 0.40 seconds per epoch in Phase 1, compared with 2.39 seconds per epoch for direct MILP training. The two-phase run took 8.43 seconds versus 47.71 seconds for direct MILP training (5.66× speedup) and reached a best validation regret of 16.82% versus 23.43%. This small profiling experiment does not establish end-to-end daily retraining feasibility at production scale; that claim requires evaluation on representative data volumes, hardware, and retraining workloads.

8 Conclusion and Future Research Directions

This paper develops a **Decision-Focused Learning (DFL)** framework for the Multi-Period Perishable Inventory-Routing Problem (PIRP) under asymmetric spoilage and stockout penalties. We present an LP recourse formulation and derive recourse subgradients with respect to replenishment decisions (Proposition 1); separately, we define a custom stop-gradient update for predicted demand (Eq. (44)), which is not an exact subgradient of the MILP task loss. We also describe RHS interior-point sensitivity and perturbed optimization approaches. Theorem 1 establishes complete recourse and boundedness on the stated deterministic feasible set under its assumptions. Algorithm 2 specifies LP-relaxation warmup followed by MILP evaluation and stop-gradient fine-tuning.

The experiments compare 14 models across five seeds, three spatial topologies, and an 81-configuration sensitivity grid. On the primary benchmark, `Two-Phase_DFL` has the lowest reported normalized decision regret ($22.28\% \pm 0.08\%$) and mean operational cost ($\$2,143.60 \pm \$1.33$), with the reported paired tests indicating lower regret than each of the 13 baselines after Holm adjustment ($p_{\text{Holm}} < 0.05$). The sensitivity results also show that Two-Phase DFL is not uniformly best: at the tested tight-capacity setting ($\gamma = 0.8$), LightGBM has lower regret (5.60% versus 6.91%). The capacity and quantile analyses are specific to the tested benchmark and do not establish a general causal mechanism. In a separate solver-profiling experiment, the two-phase schedule took 8.43 s versus 47.71 s for direct MILP training; this result should be interpreted within that profiling setup rather than as evidence of production-scale retraining time.

Future research directions include:

(i) Extending the framework to multi-echelon cold-chain distribution networks with multiple intermediate temperature-controlled cross-docks;

(ii) Incorporating stochastic travel times and real-time dynamic rerouting into the decision-focused learning loop;

(iii) Exploring distributionally robust decision-focused learning (DRO-DFL) under Wasserstein ambiguity sets for non-stationary perishable product life cycles.

Declarations

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Conflicts of Interest: The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Ethics Approval: Not applicable. This study does not involve human participants or animal subjects.

Data and Code Availability: All simulation code, experimental pipeline scripts, preprocessed benchmark instances, and evaluation datasets supporting the findings of this study are available in the project repository at https://github.com/DeckardShaw31/Two-Phase-DFL.

Declaration of Generative AI and AI-assisted technologies in the writing process: During the preparation of this work, the authors used generative AI tools in order to assist with code optimization, data formatting, and grammar refinement. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.